# Application Notes on Optics
# - with lightly theoretical approach

2011 April / May

Chih-Han Lin 林致翰

# Contents

Introduction

Wave, Geometric & Gaussain optics

Polarization

Terinology &  general optical components

Beam alignment

Caculus of beam propagation : ray matrix

Investigation on Gaussian beam propagation

to be continued...

# History of Light

Genesis 1-3

And God said, "Let there be light" and there was light.

Hook v.s. Newton 波動 or 粒子 ?

Thomas Young 的干涉實驗說光是波動

Hertz 的實驗表明光即 Maxwell 理論中的電磁波

Einstein 對光電效應的解釋則表明光的粒子性

Taylor 單光子干涉實驗, 波粒二象性

visible light



# Optics

λ ->0

波動光學 <-----高斯光學-----> 幾何光學

| Arbitary wavefront | Guassian wave | Plane Wave |
| --- | --- | --- |
| 折射、反射 | 近軸近似 | 折射、反射 |
| 繞射、偏振 | 繞射、偏振 | 成像系統 |
| | Matrix Calculation | Matrix Calculation |

電磁場 -> function of space & time

折射、反射

# Optics

## Wave Optics

$$\nabla^2\mathbf{E} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\mathbf{E} = 0 \quad \text{(free space)}$$

$$\mathbf{E}(\mathbf{r},t) = \tilde{\mathbf{E}}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$$

vector E-field

## Geometric Optics

$$\nabla^2 U(\mathbf{r}) + \frac{\omega^2}{n^2c^2}U(\mathbf{r}) = 0$$

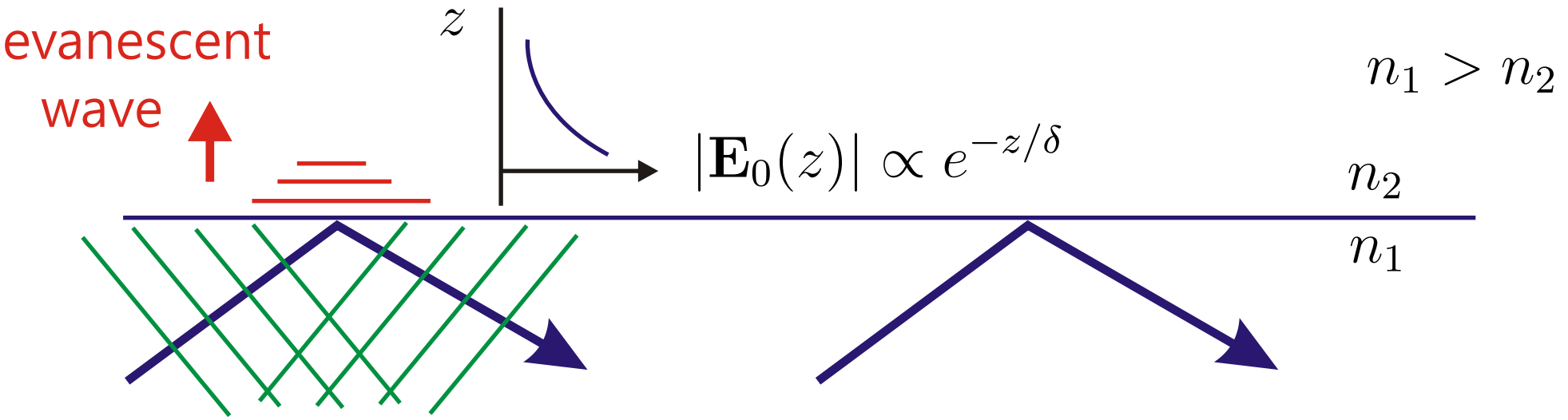
$$E(r,t) = U(\mathbf{r})e^{i\omega t} = A(\mathbf{r})e^{iS(\mathbf{r})}e^{i\omega t}$$

$$|\nabla S(\mathbf{r})|^2 \approx n^2$$

Scalar E-field

no evanescent wave

# Total Reflection

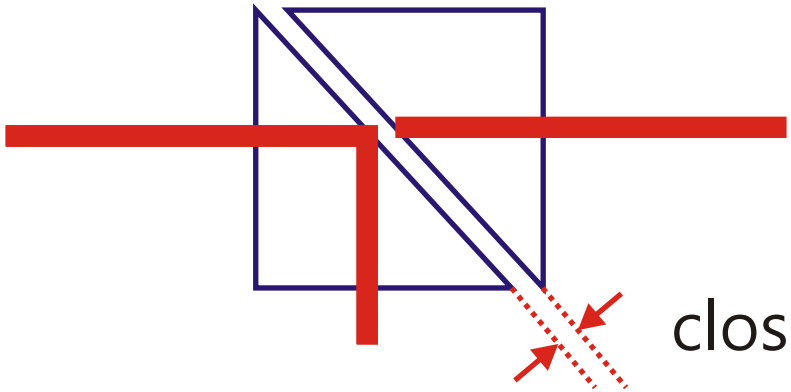



wave optics (plane wave)

geometric optics

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\theta_i > \theta_c = \sin^{-1}\left(\frac{n_2}{N_1}\right)$$



close to skin depth

# Wave Optics

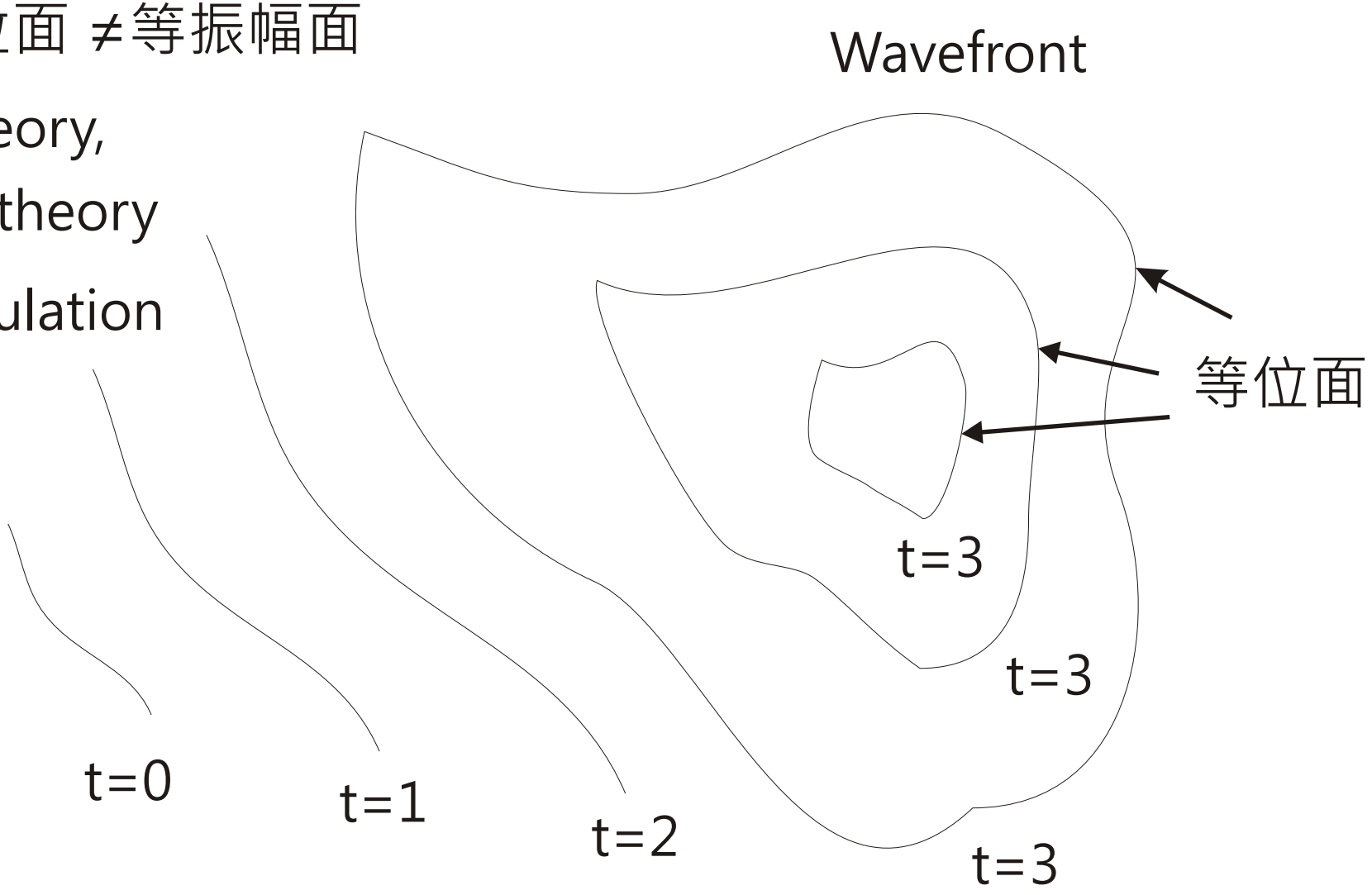
Global property

等（相）位面 ≠等振幅面

Maxwell theory,

Diffraction theory

Direct Calculation for "field"



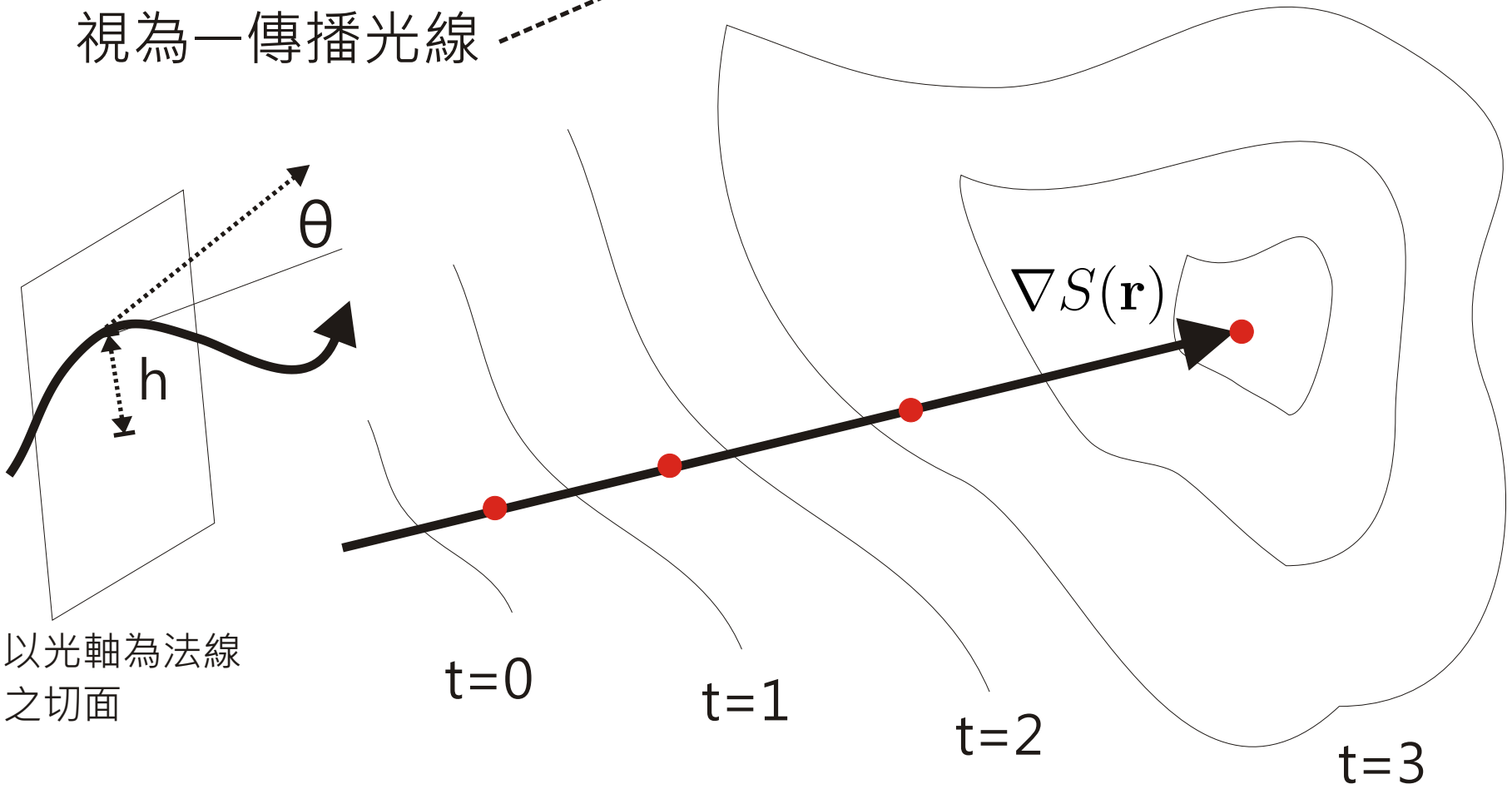
# Geometric Optics

Local property

不一定是直線，隨 n(x,y,z)而定 $|\nabla S(\mathbf{r})|^2 \approx n^2$

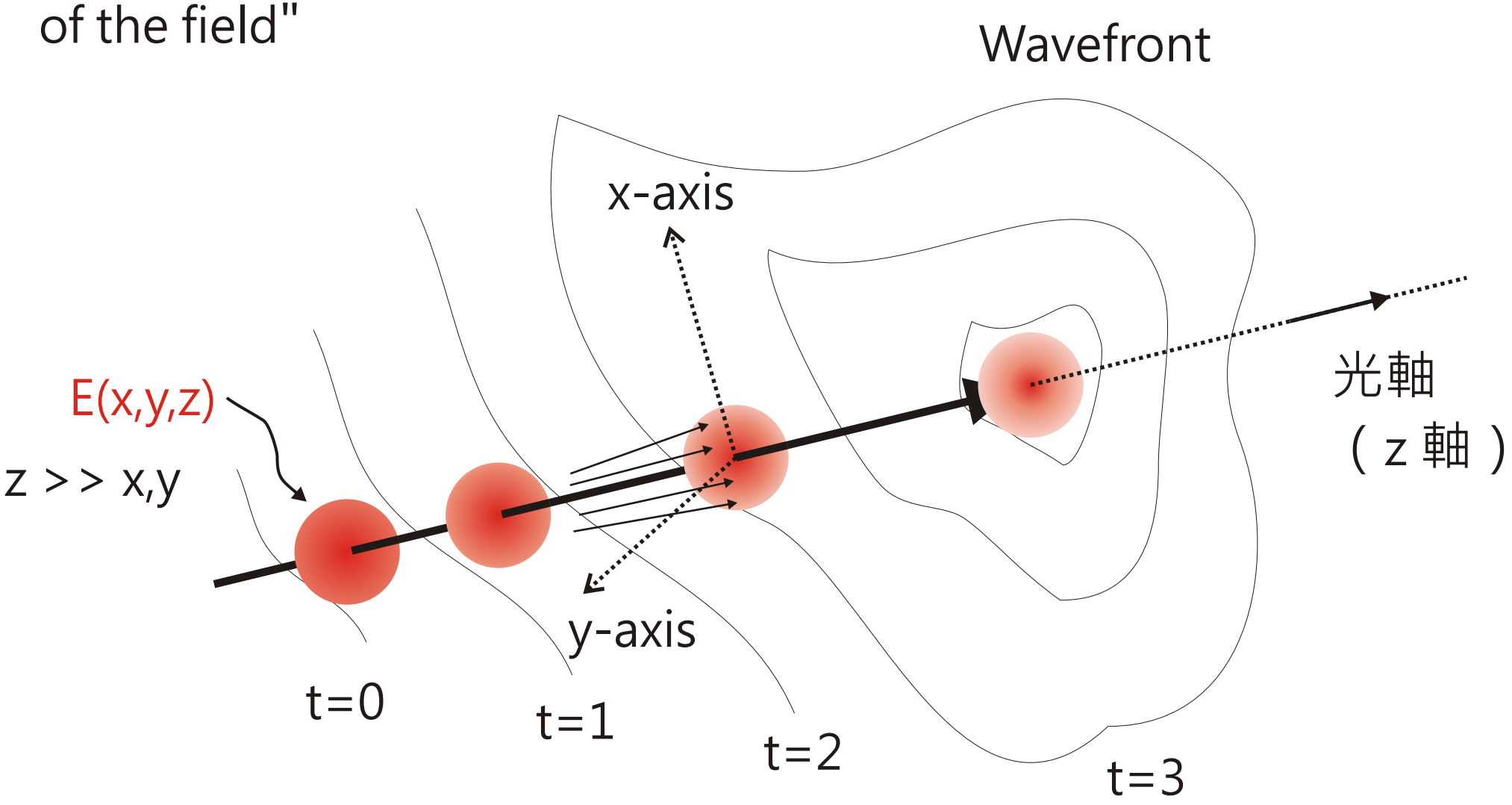
等（相）位面法線
視為一傳播光線

Wavefront



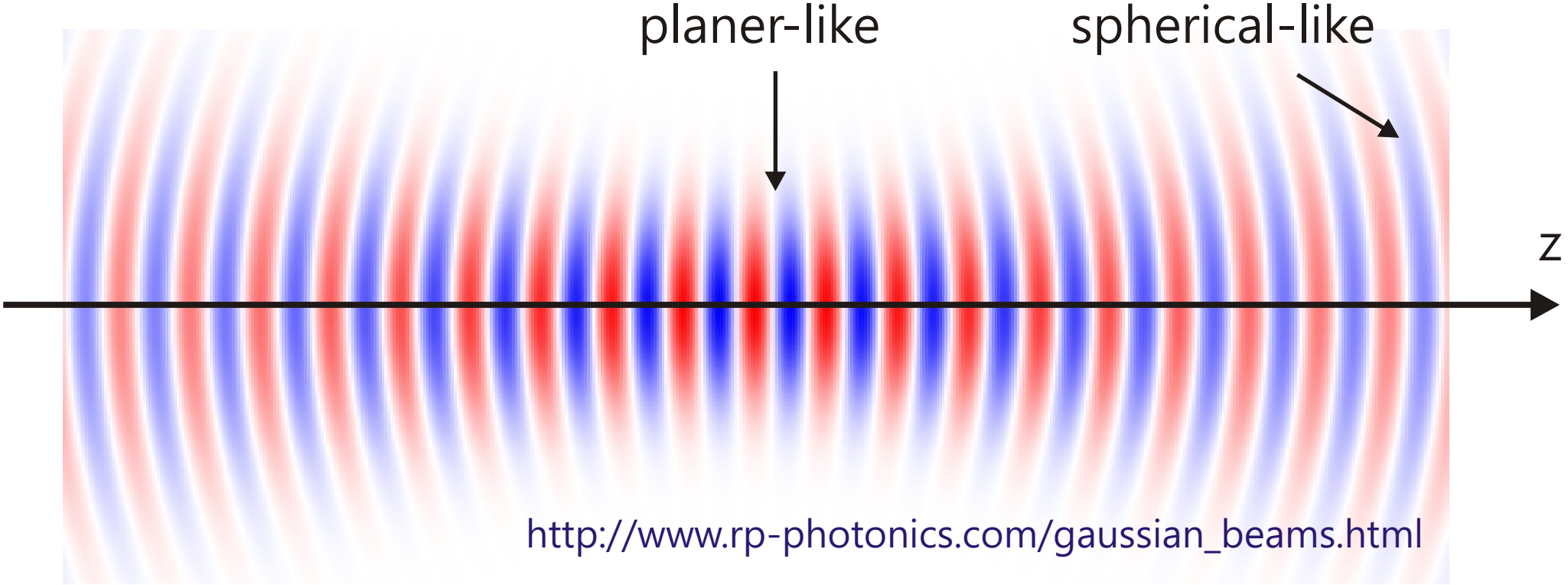
# Gaussian Optics

Calculation for "paraxial approximation of the field"



# Gaussian Optics

Electric field distribution around the focus of a Gaussian beam



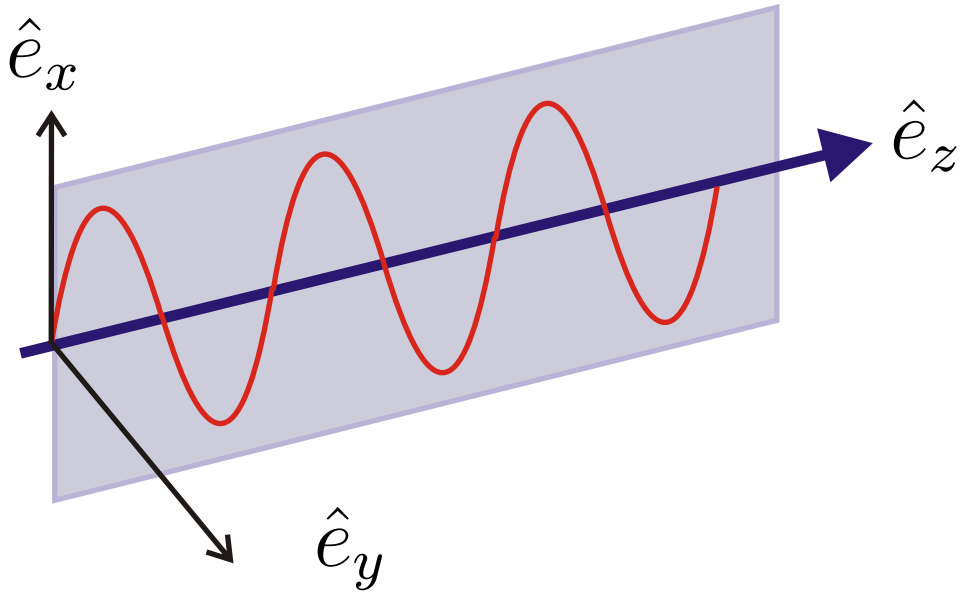

phase=0 ->red,  phase=180 deg -> blue

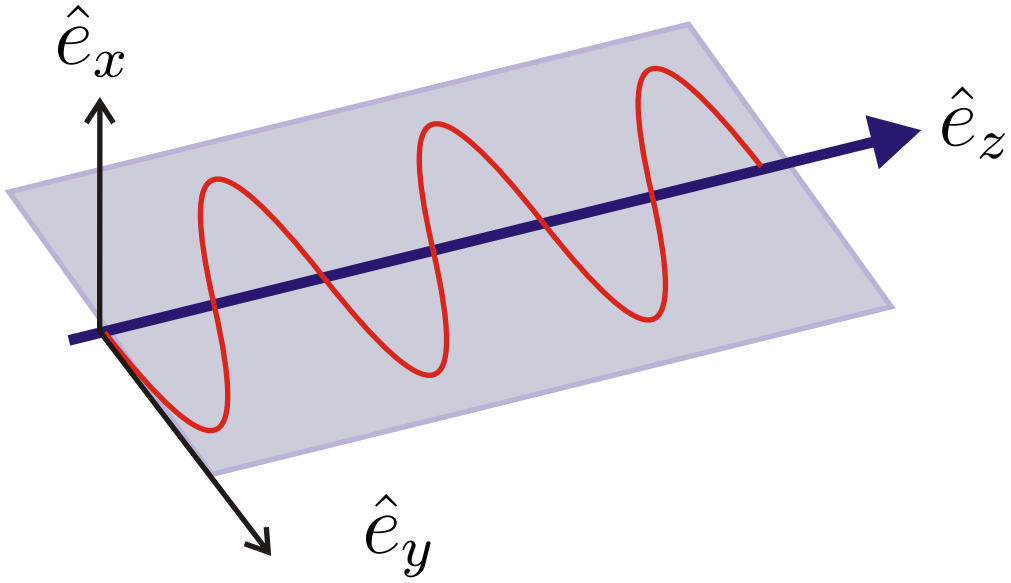
# Polarization (Polarisation)

$$\mathbf{k} \cdot \mathbf{E} = 0 \qquad \mathbf{k} = k\hat{e}_z$$

$$\mathbf{E}(z,t) = E_x(z,t)\hat{e}_x + E_y(z,t)\hat{e}_y$$

$$E_x(z,t) = E_{x_0} e^{i(kz-\omega t+\theta_1)} \qquad E_y(z,t) = E_{y_0} e^{i(kz-\omega t+\theta_2)}$$

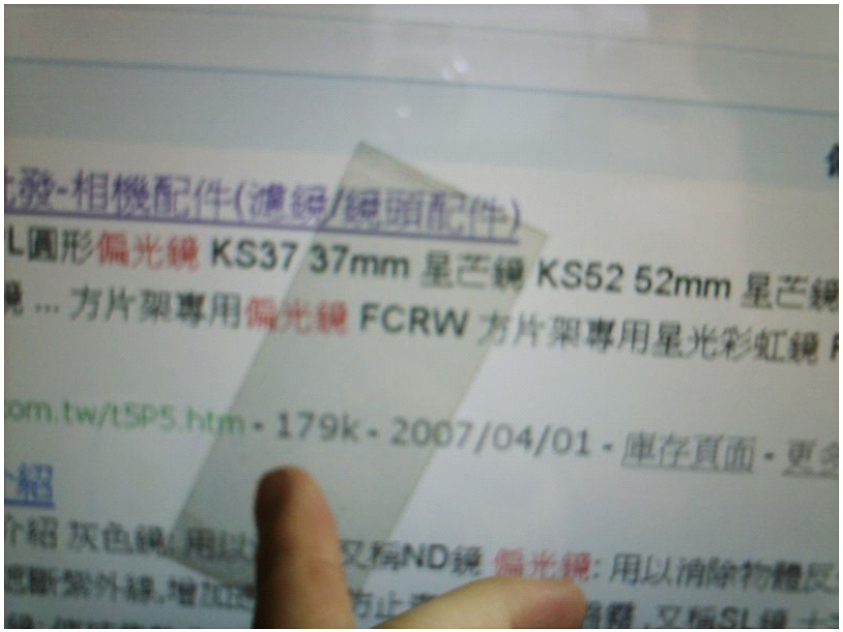


# Polarization (Polarisation)

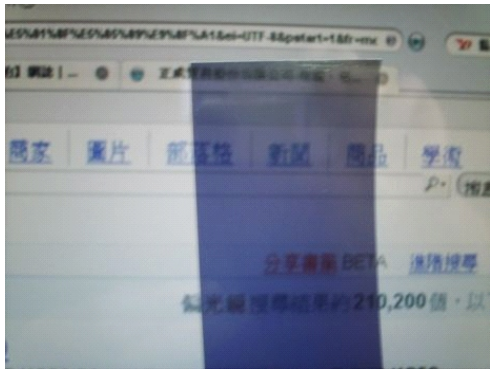
ex. linear polarized light from LCD



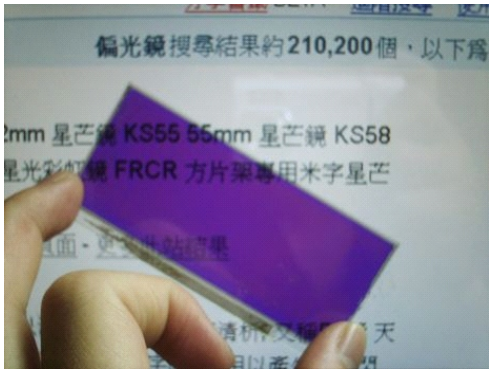



linear polarizer

# Polarization (Polarisation)

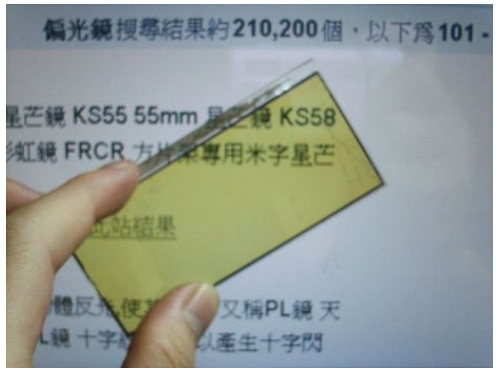














linear polarizer + a layer of medium or liquid crystal

Can you explain what you see?

# Polarization (Polarisation)



with linear polarizer



without linear polarizer

# Polarization (Polarisation)

without linear polarizer

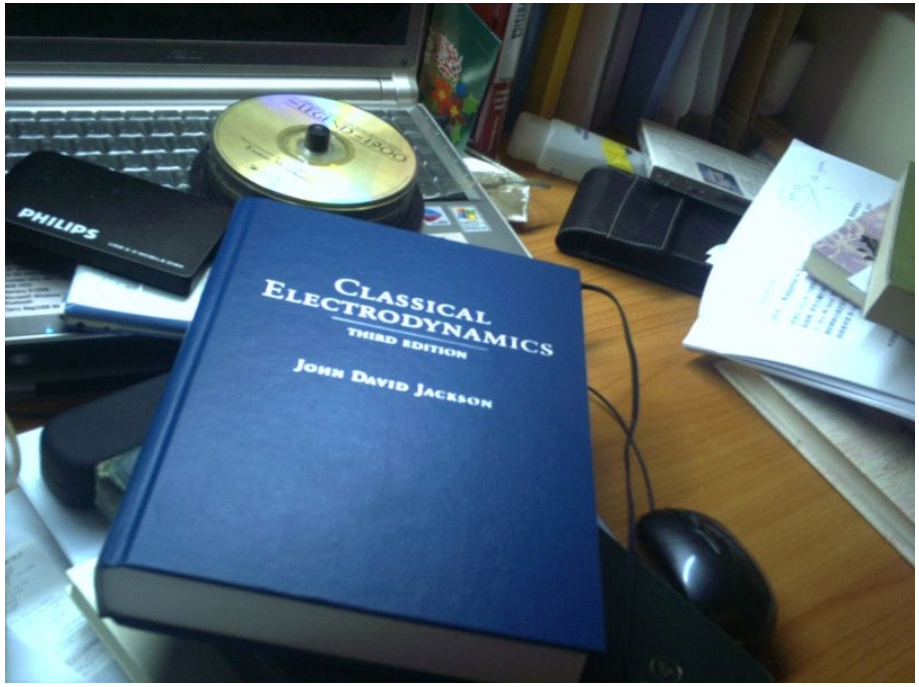




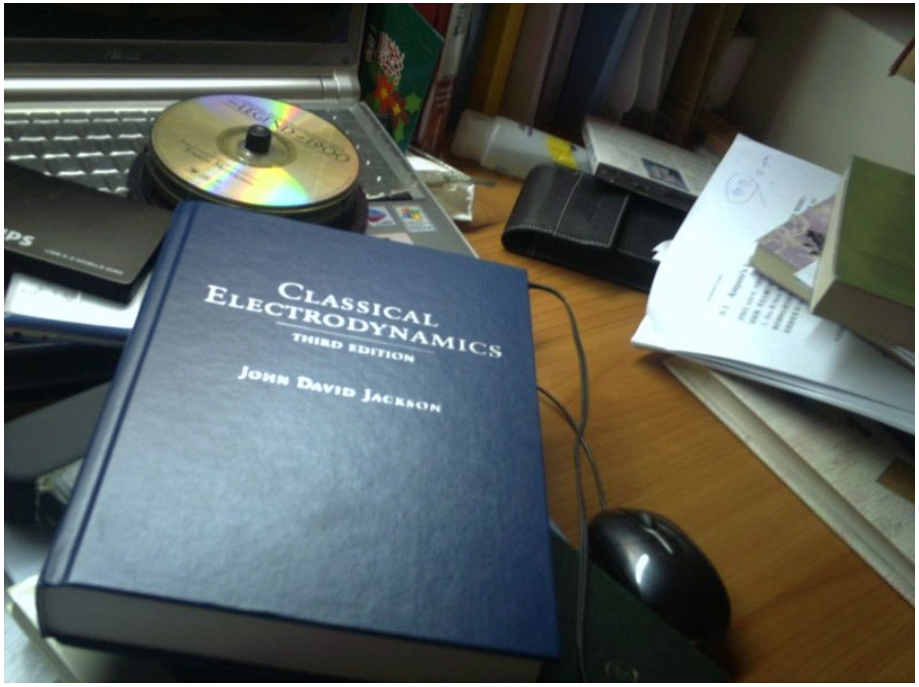
with linear polarizer

# Polarization (Polarisation)

non-metallic surface scattering




with linear polarizer




without linear polarizer

# Polarization (Polarisation)



with linear polarizer



without
linear polarizer

# Polarization (Polarisation)



http://www.polarization.com/sky/sky.html

This fisheye photograph (with LP) of the full sky at sunset

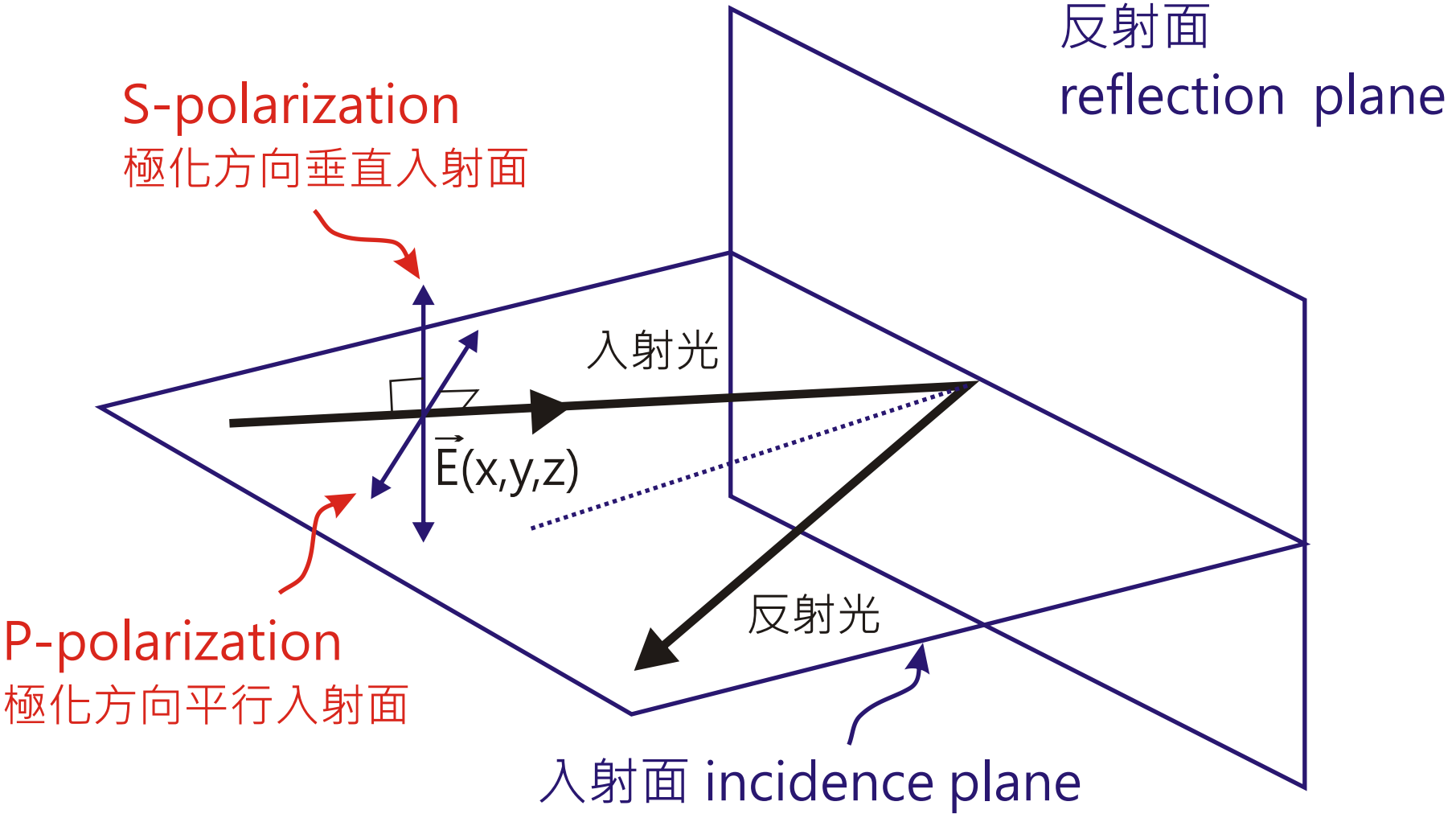
blue light is less polarized than other colors in skylight.




『桂魄飛來光射處，冷浸一天秋碧』

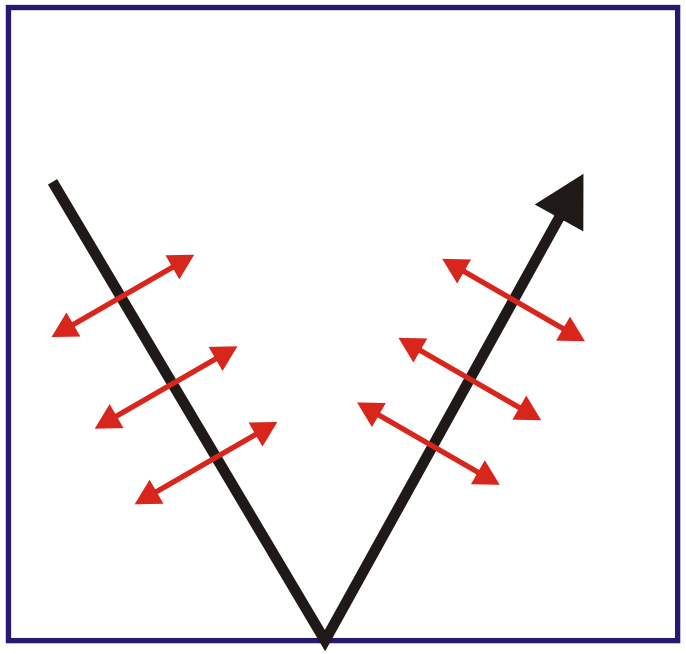
『今日北池遊。漾漾輕舟。波光瀲灩柳條柔。
如此春來春又去，白了人頭。 』

# Terminology

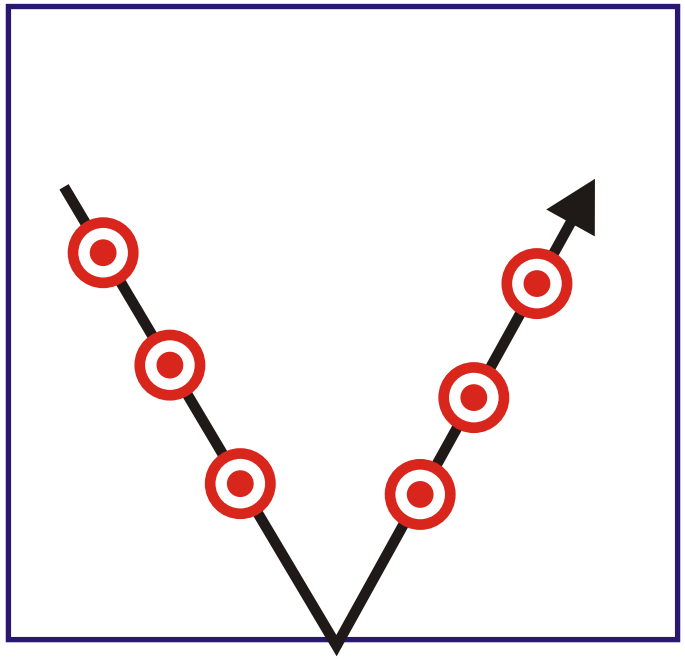


# Terminology

P-polarization

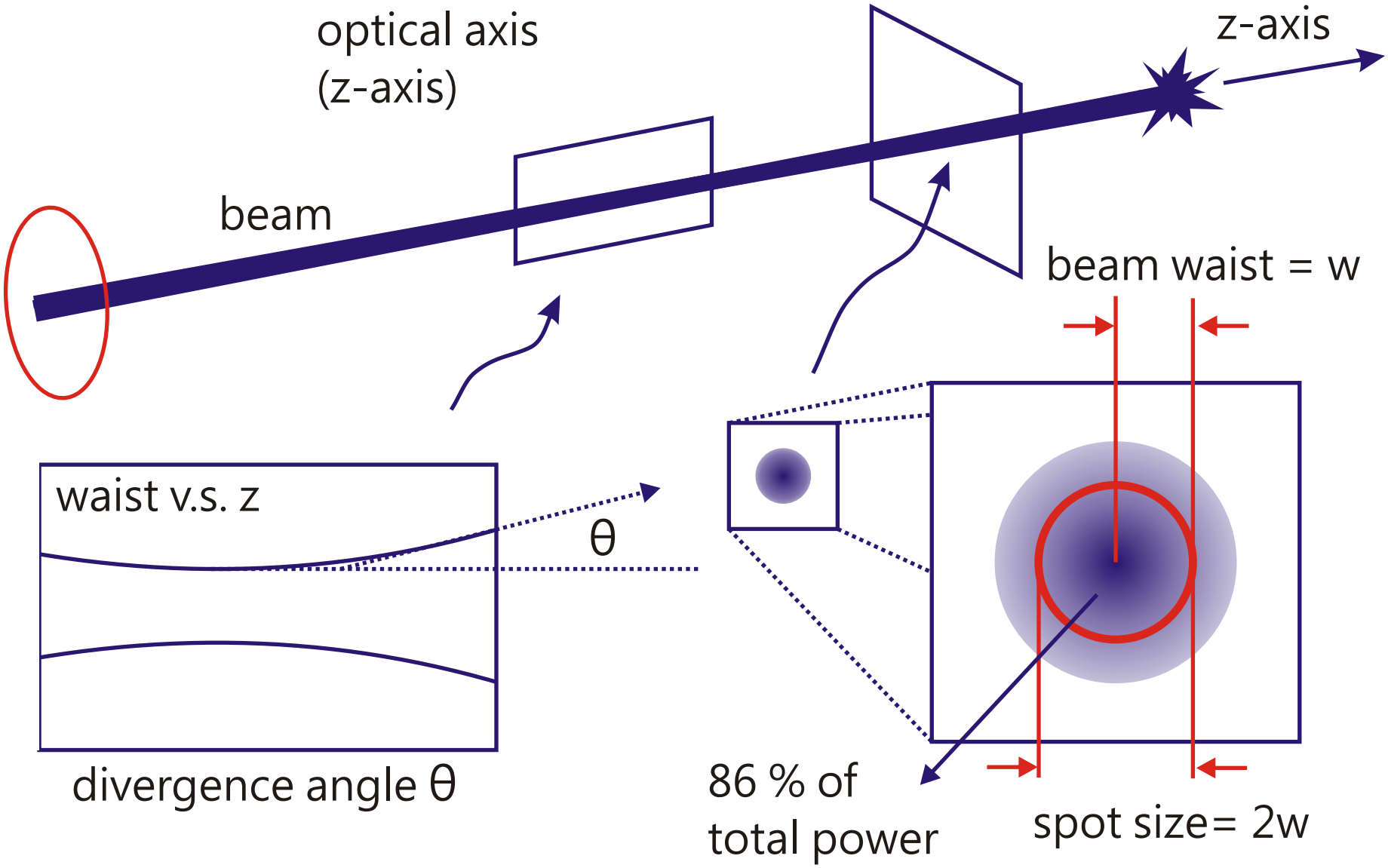


S-polarization



入射面 incidence plane

# Terminology



# Terminology

## C.A. (clear aperture)

for lens, typically 90%
of device diameter
f-number: f / C.A

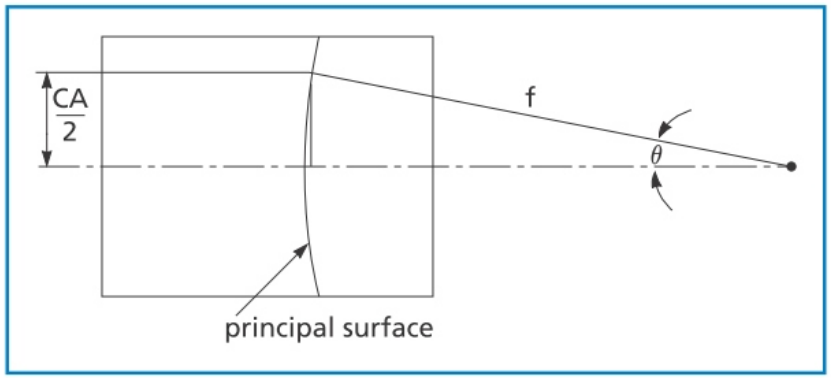
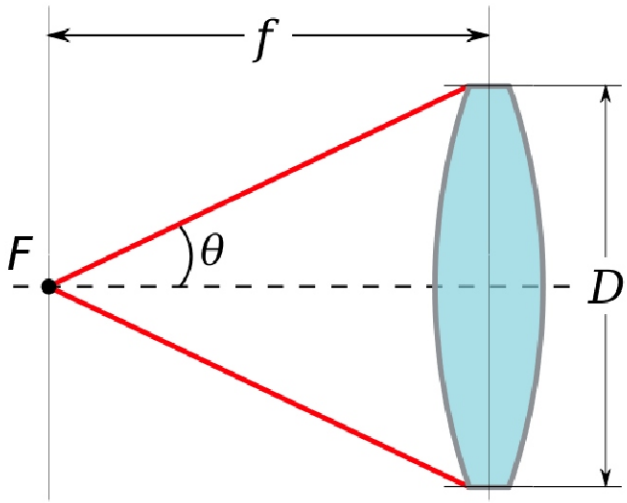



Figure 1.5 **F-number and numerical aperture**

## N.A. (numerical aperture)

$$N.A = n \sin\theta \fallingdotseq \frac{C.A.}{2f}$$



# Terminology

M (magnification, 10X, 20X, 40X ...)

differ by manufacturer

$$M = \frac{254\ \text{mm (or 187 mm, tube length)}}{f\ \text{(focal length)}}$$

ex. N.A.=0.85 40 X objective lens

f=187 mm / 40 = 4.675 mm

C.A.≒( 2f ) N.A.=2 x 4.675 mm x 0.85

=7.9 mm




http://www.olympus-global.com/en/corc/history/micro/image/uis_umplan_l.jpg

# Terminology

## Transversal mode (spatial mode)

characteristic of spatial domain

single mode -> E(x,y) along z-axis  is guassian distribution

## longitudinal mode

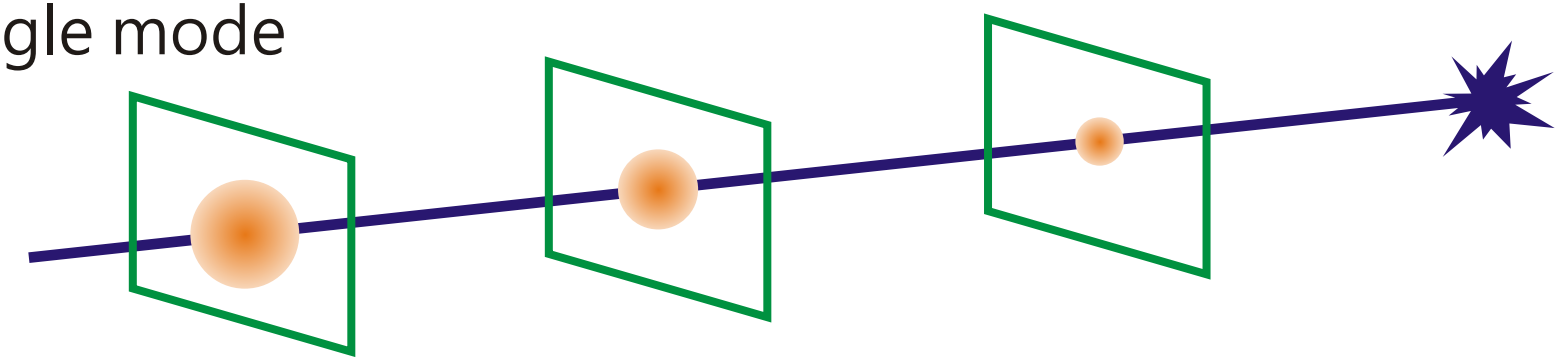
characteristic of frequecy domain

single mode -> single frequency output

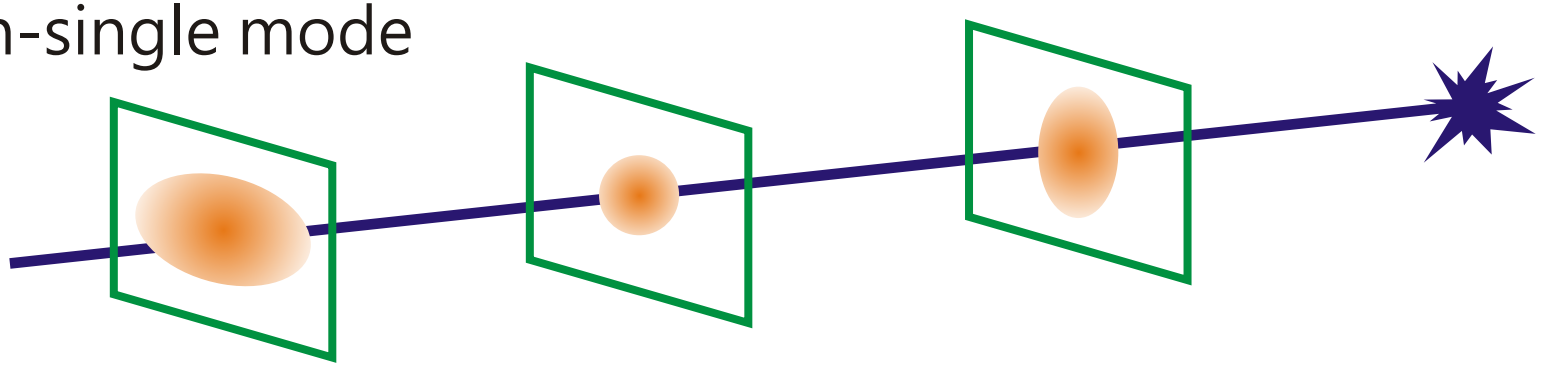
# Terminology

## Transversal mode (spatial mode)

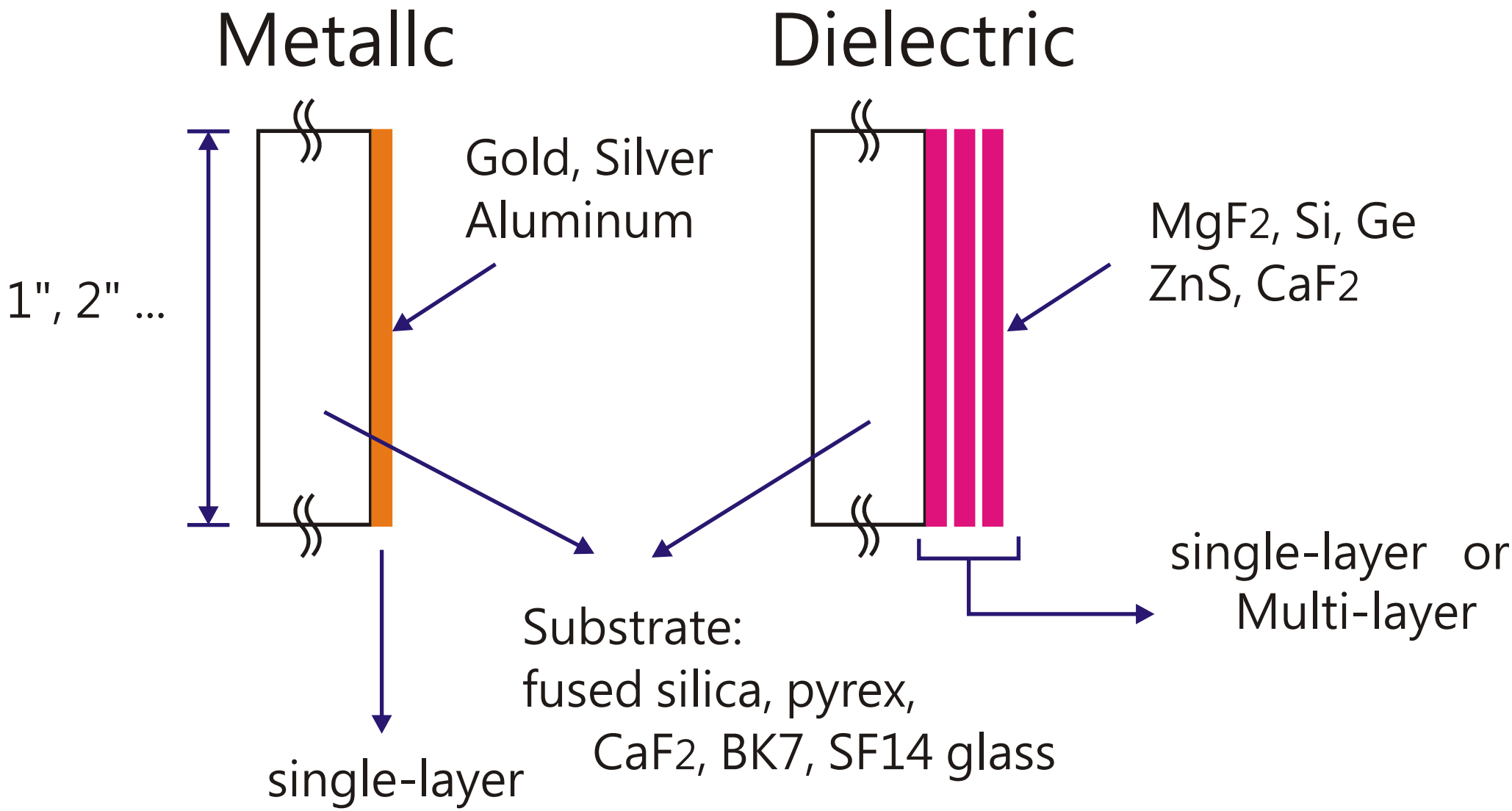
single mode



non-single mode



# Mirrors


Metallc
Dielectric
1", 2" ...
Gold, Silver
Aluminum
MgF2, Si, Ge
ZnS, CaF2
single-layer or
Multi-layer
Substrate:
fused silica, pyrex,
CaF2, BK7, SF14 glass
single-layer

# Mirrors

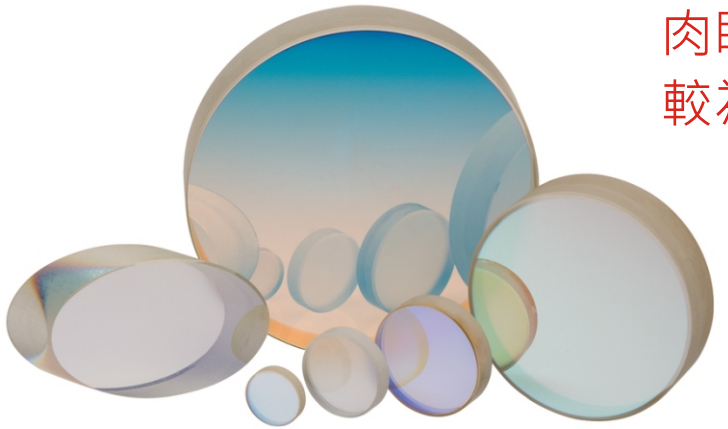
## Metallic



pic. from Newport Website

對於極化，入射角較不敏感，
反射頻寬較大

## Dielectric

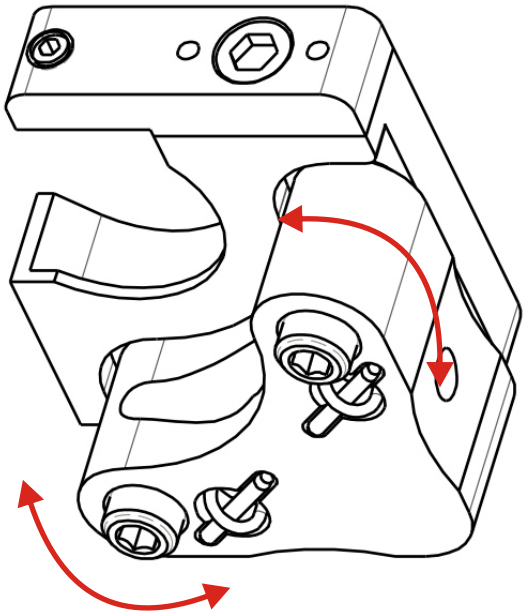


肉眼觀察
較為透明

pic. from Newport Website

破壞閥值較高，可以透過鍍模結構調控反射頻譜
ex: BBAR (broadband anti-reflection) coating, V-coating

# Mirror Mounts

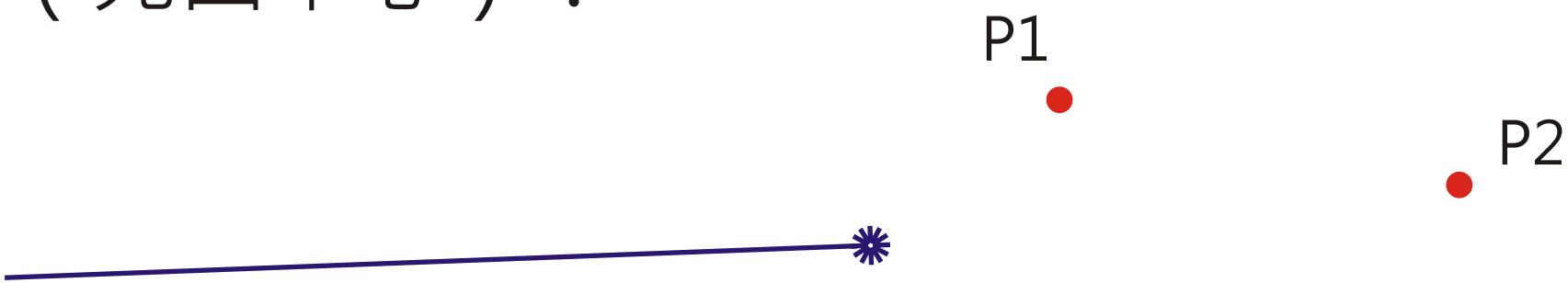





pic. from Newport Website

# 如何對光

平面上有兩點（兩個光圈），一束方向固定的雷射光，如何使用反射鏡將光束導引過兩點（光圈中心）？

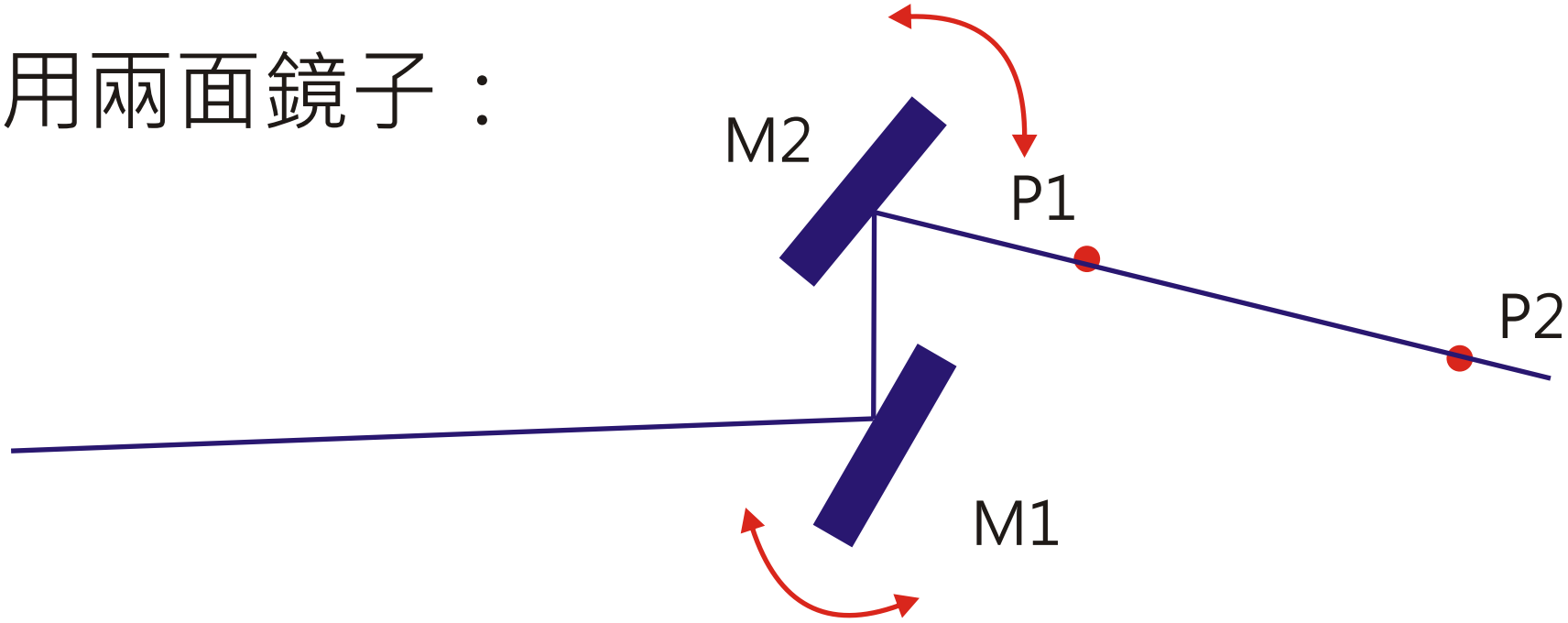



hint: 要找到兩個調整參數

hint: 調整的方法要能收斂

# 如何對光

使用兩面鏡子：




M2 先大置放在 P1,P2 兩點延伸處，大致轉動 M1,M2 的角度使光通過 P1, P2 (光圈全開)。

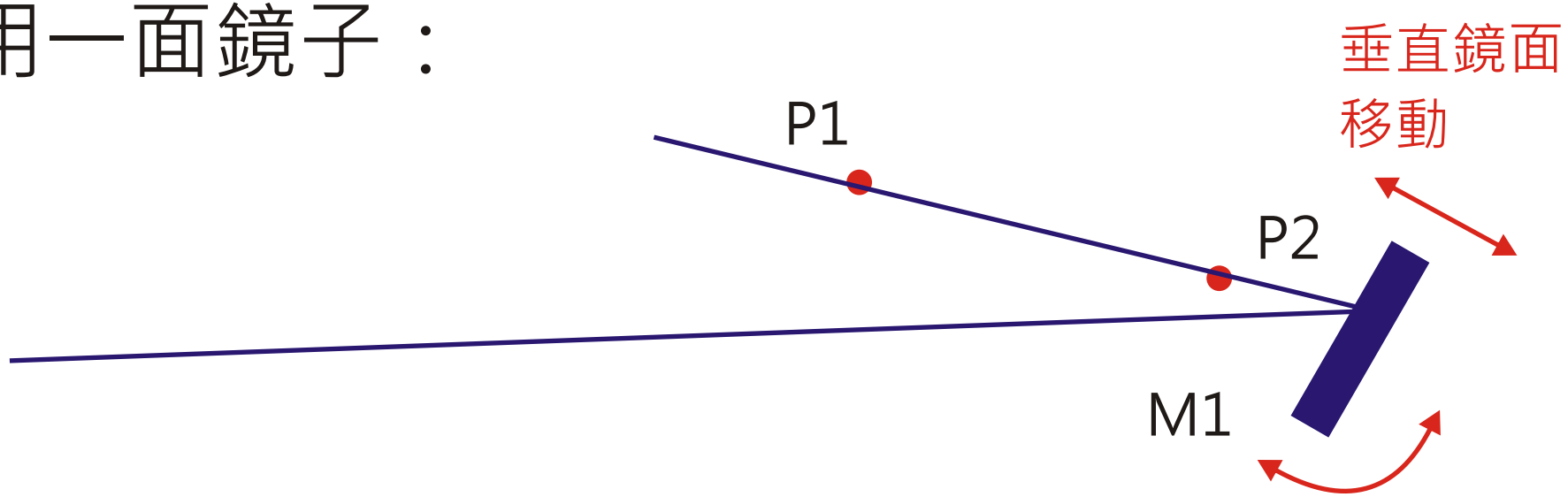
step A: 打開 P1 光圈，細調節 M2 使光對準 P2 中心

step B: 關閉 P1 光圈，細調節 M1 使光對準 P1 中心

重複執行 step A,B 直至收斂

# 如何對光

## 使用一面鏡子：




M2 先大置放在 P1,P2 兩點延伸處

step A: 打開 P2 光圈，細調節(轉動) M1 使光對準 P1 中心

step B: 關閉 P1 光圈，移動(不平行鏡面即可) M1 使光對準 P2 中心

重複執行 step A,B 直至收斂

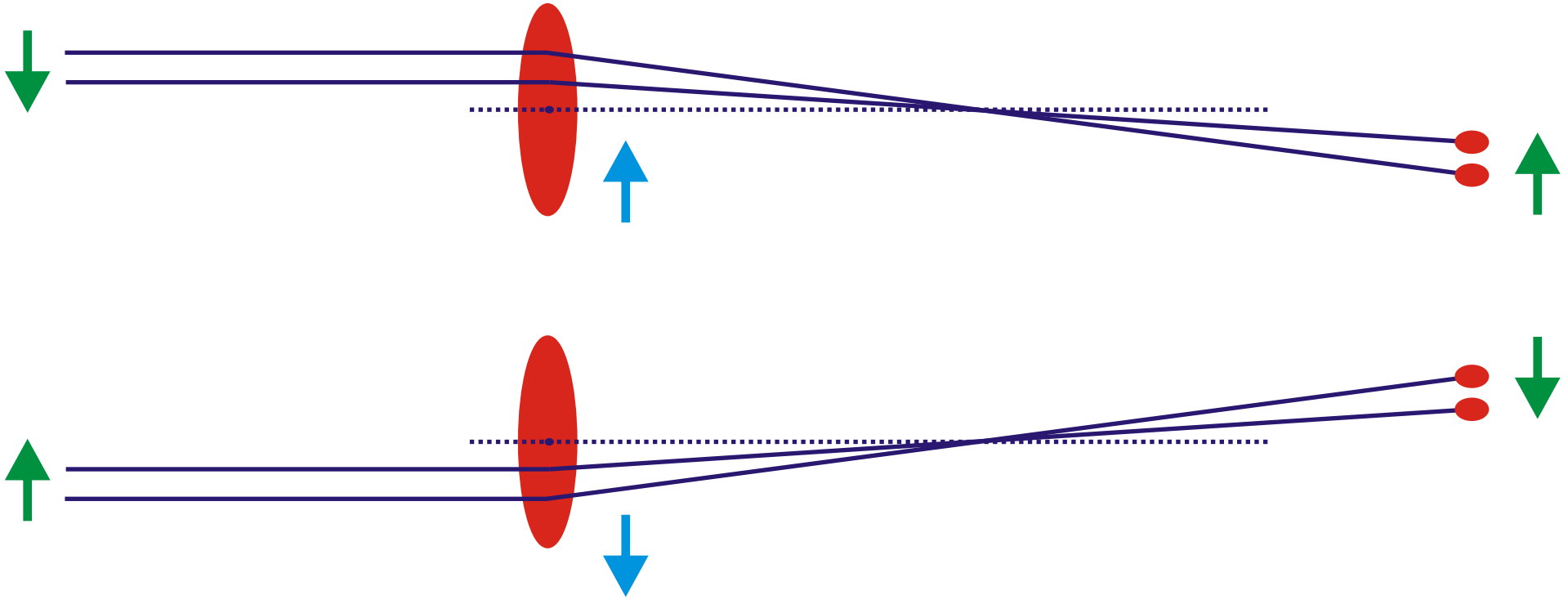
最好架設簡單滑軌方便移動（不破壞角度）

# 如何對透鏡（固定透鏡）

1. 確定光束通過鏡心
   未通過鏡心則輸出光束不在光軸上

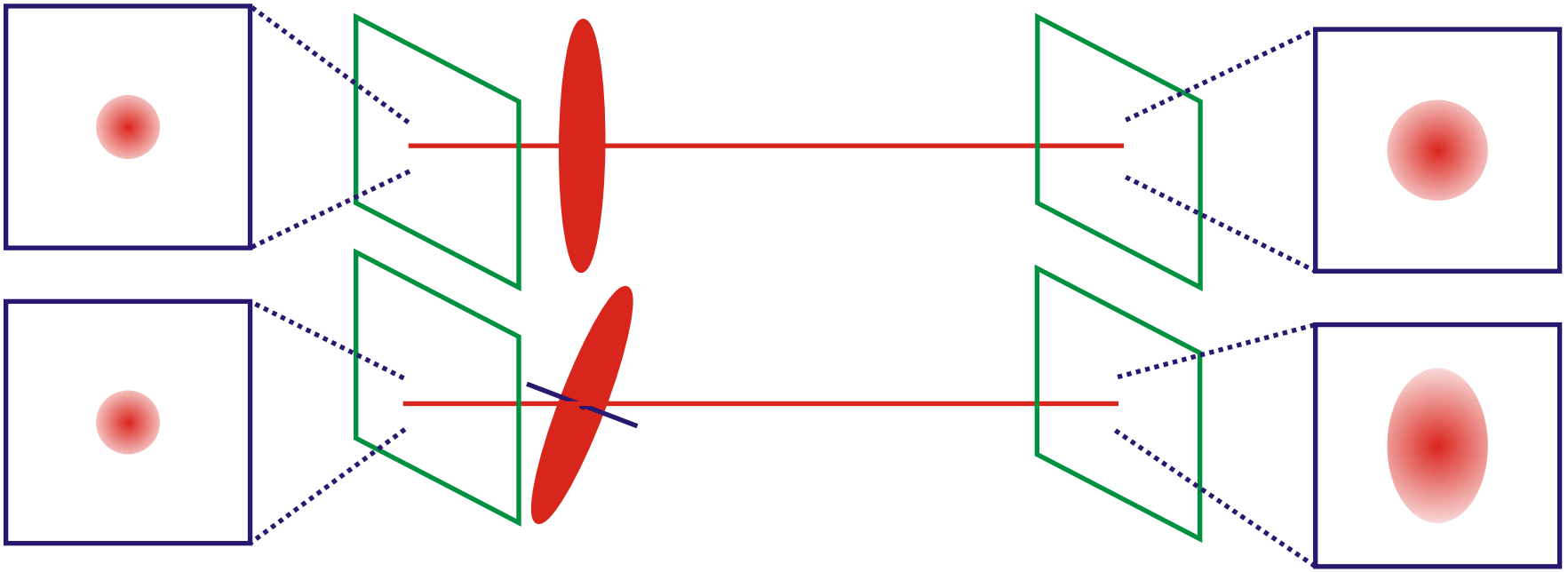
2. 確定光束正向入射
   未正向入射鏡心則輸出光束形狀會變（橢圓）

# 如何對透鏡（固定透鏡）



未放上透鏡時，在遠端光束處作記號（在屏幕上標記），放上透鏡後
若光點偏上（左）則透鏡往下（右）移動
若光點偏下（右）則透鏡往上（左）移動

# 如何對透鏡（固定透鏡）

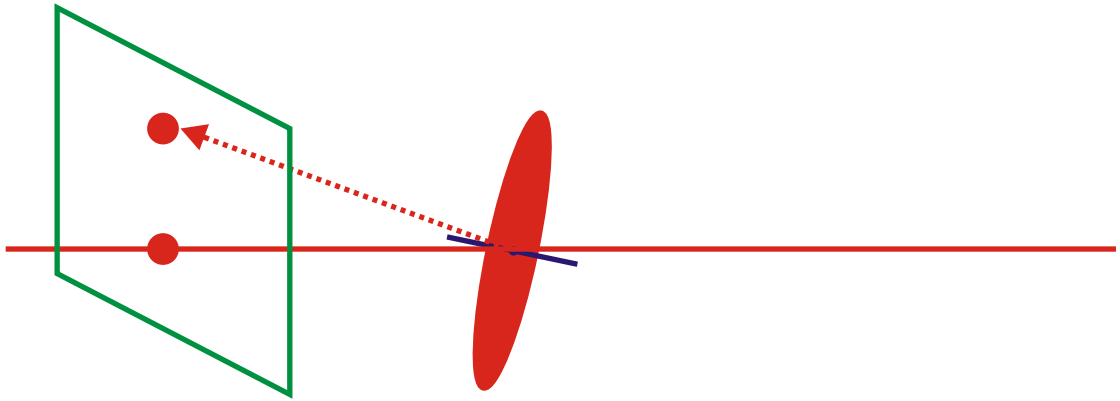


未放上透鏡時，在遠端光束處放上同心圓靶，放上透鏡後
若光點形狀上下變形，則透鏡上下轉動
若光點形狀左右變形，則透鏡左右轉動

使用電腦繪製同心圓靶紙備用

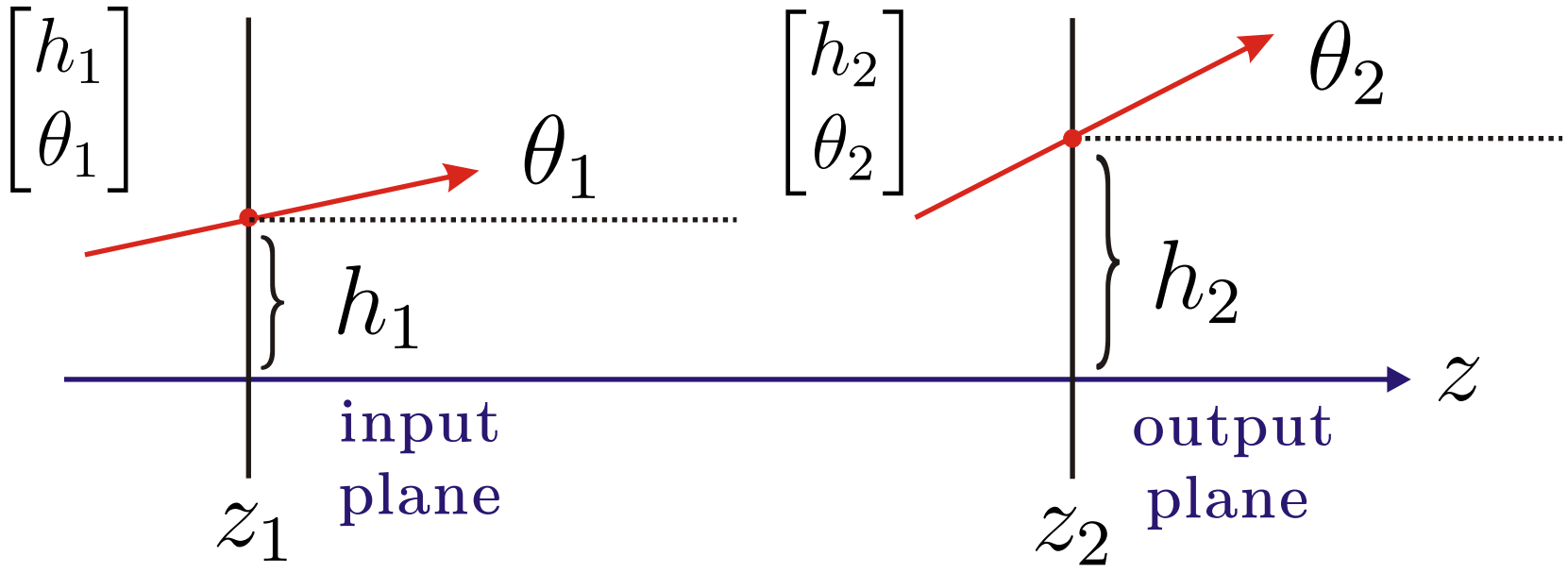


# 如何對透鏡（固定透鏡）



若可看到透鏡正面微弱反射光，也可用其校準透鏡角度

# Ray-transfer optics



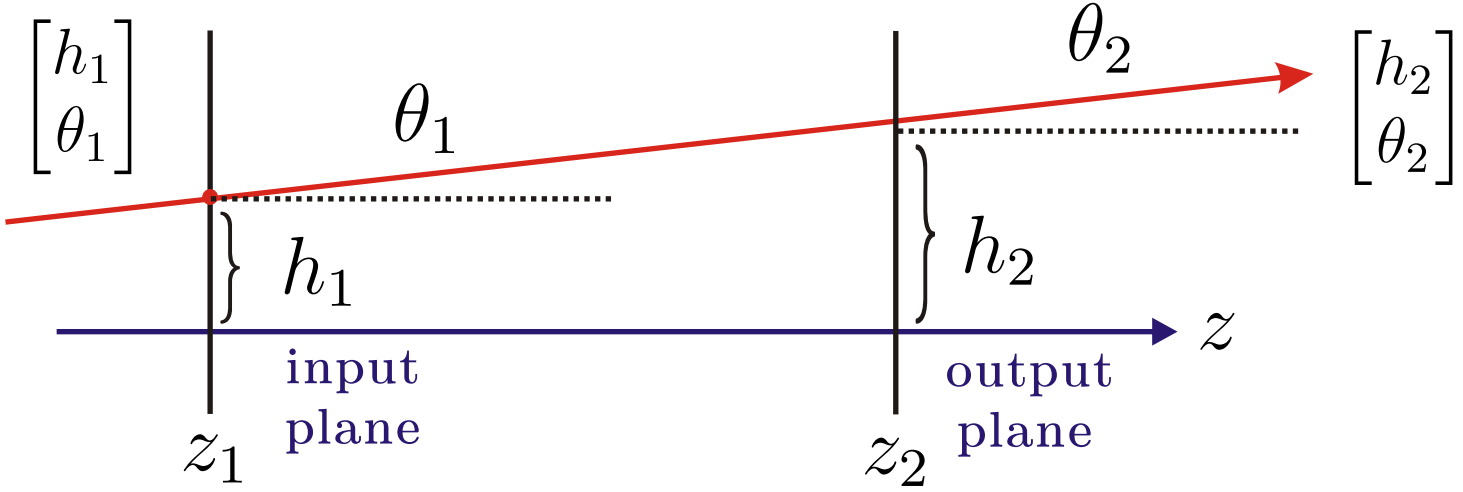

$$h_2 = Ah_1 + B\theta_1,$$
$$\theta_2 = Ch_1 + D\theta_1.$$

$$\begin{bmatrix} h_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} h_1 \\ \theta_1 \end{bmatrix} = \mathbf{M} \begin{bmatrix} h_1 \\ \theta_1 \end{bmatrix}$$

with paraxial approximation $\sin\theta \simeq \theta$

# Ray-transfer optics ex. free space propagation

$$d = z_2 - z_1$$

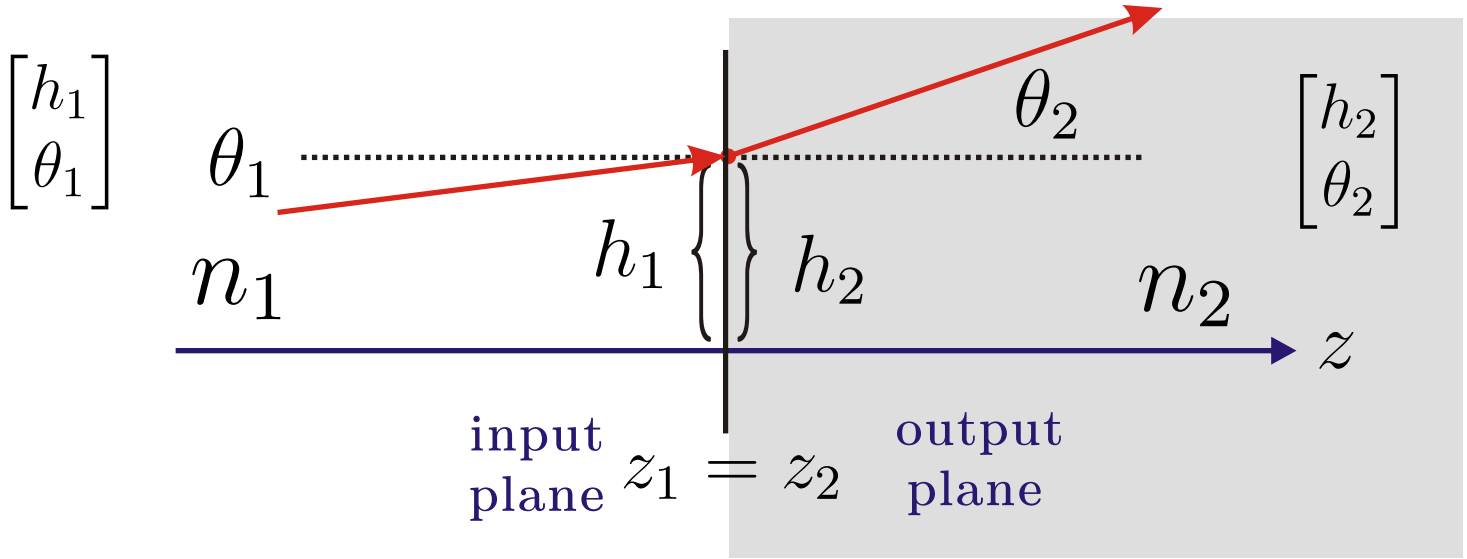



$$h_2 \simeq h_1 + d\sin\theta_1 \simeq h_1 + d\theta_1$$

$$\theta_2 = \theta_1$$

$$\mathbf{M} = \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}$$

# Ray-transfer optics ex. refraction at planar boundary

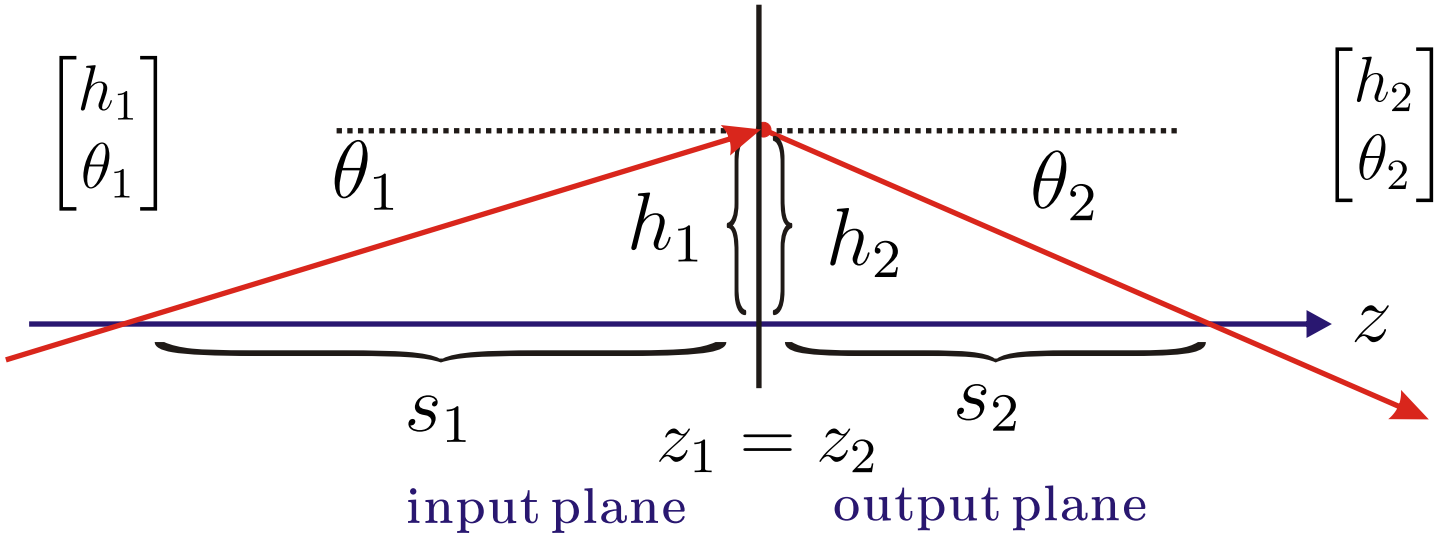



$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \rightarrow n_1\theta_1 = n_2\theta_2$$

$$h_2 = h_1$$
$$\theta_2 = (n_1/n_2)\theta_2$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & n_1/n_2 \end{bmatrix}$$

# Ray-transfer optics ex. refraction at planar boundary




$$\frac{1}{s_1} + \frac{1}{s_2} = \frac{1}{f} \qquad \frac{h_1}{s_1} + \frac{h_1}{s_2} = \frac{h_1}{f} \rightarrow \theta_1 - \theta_2 = \frac{h_1}{f}$$

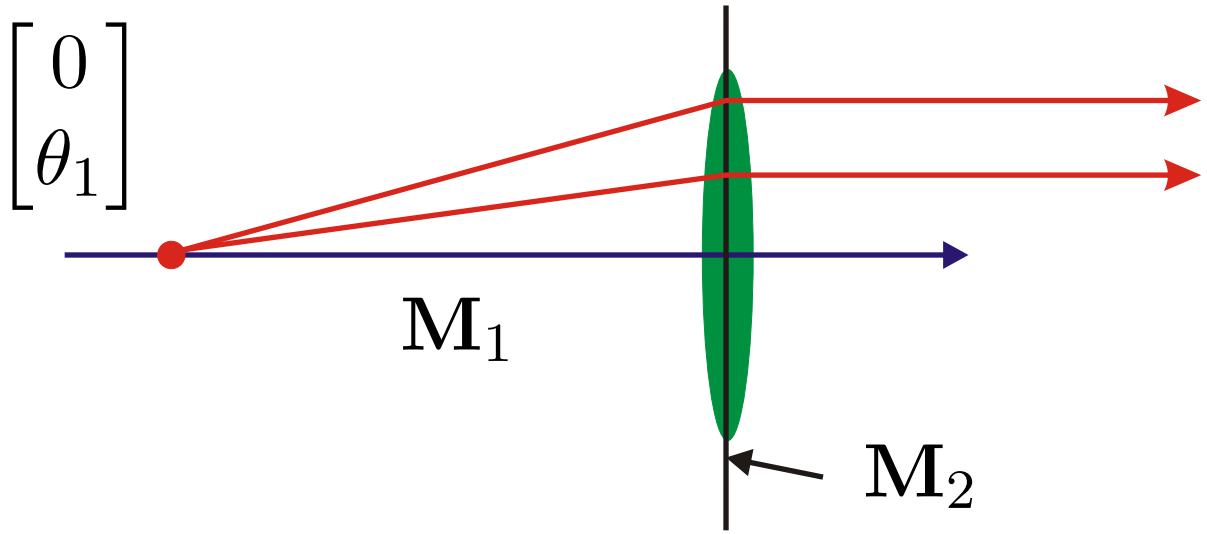
$$h_2 = h_1$$

$$\theta_2 = -\frac{h_1}{f} + \theta_1$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -f & 1 \end{bmatrix}$$

# Combination of Ray Matrix

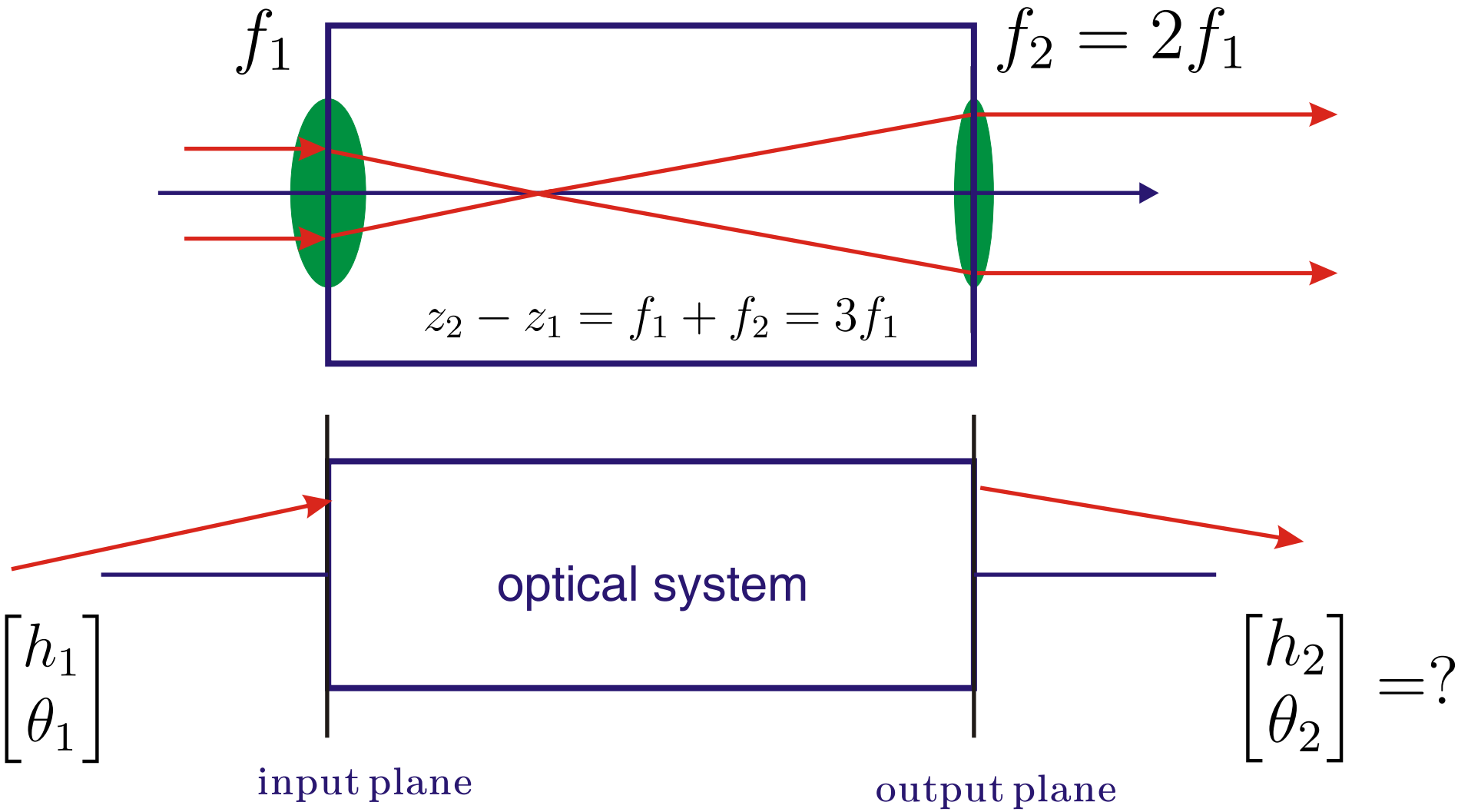
ex. collect rays from the focal spot with lens



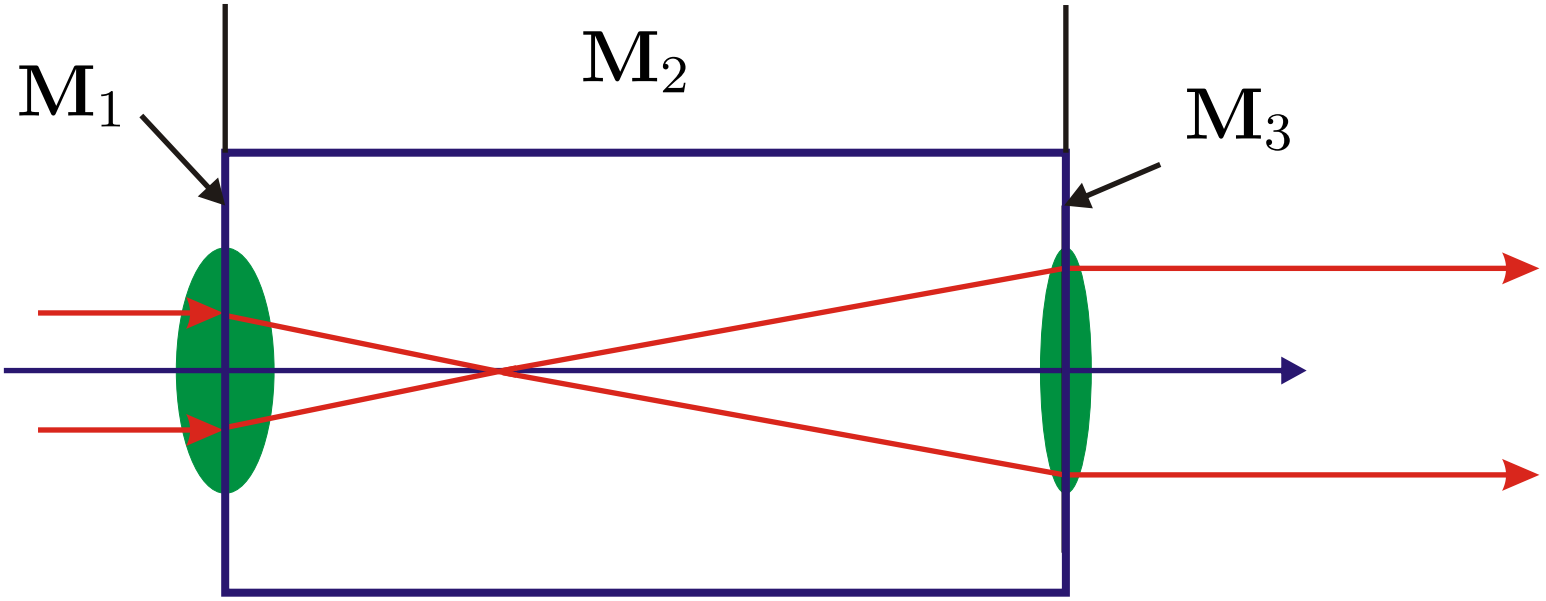

$$\mathbf{M} = \mathbf{M}_2\mathbf{M}_1 = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} 1 & f \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & f \\ -\frac{1}{f} & 0 \end{bmatrix}$$

$$h_2 = f\theta_1 \qquad \theta_2 = 0$$

# Combination of Ray Matrix ex. beam expander



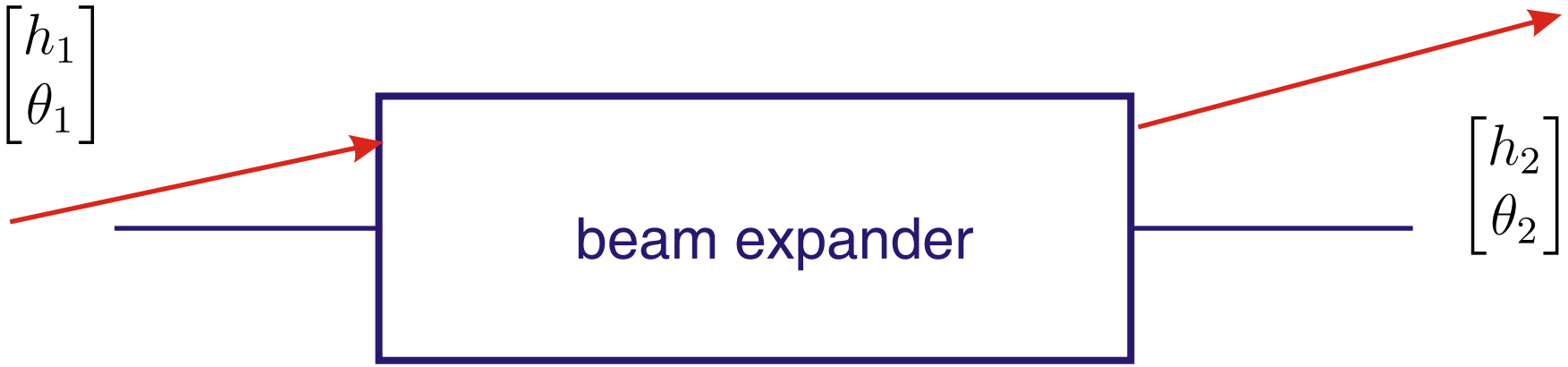
# Combination of Ray Matrix




$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_1} & 1 \end{bmatrix} \quad \mathbf{M}_2 = \begin{bmatrix} 1 & 3f_1 \\ 0 & 1 \end{bmatrix} \quad \mathbf{M}_3 = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2f_1} & 1 \end{bmatrix}$$

$$\mathbf{M} = \mathbf{M}_3\mathbf{M}_2\mathbf{M}_1 = \begin{bmatrix} -2 & 3f_1 \\ 0 & -\frac{1}{2} \end{bmatrix}$$

# Combination of Ray Matrix

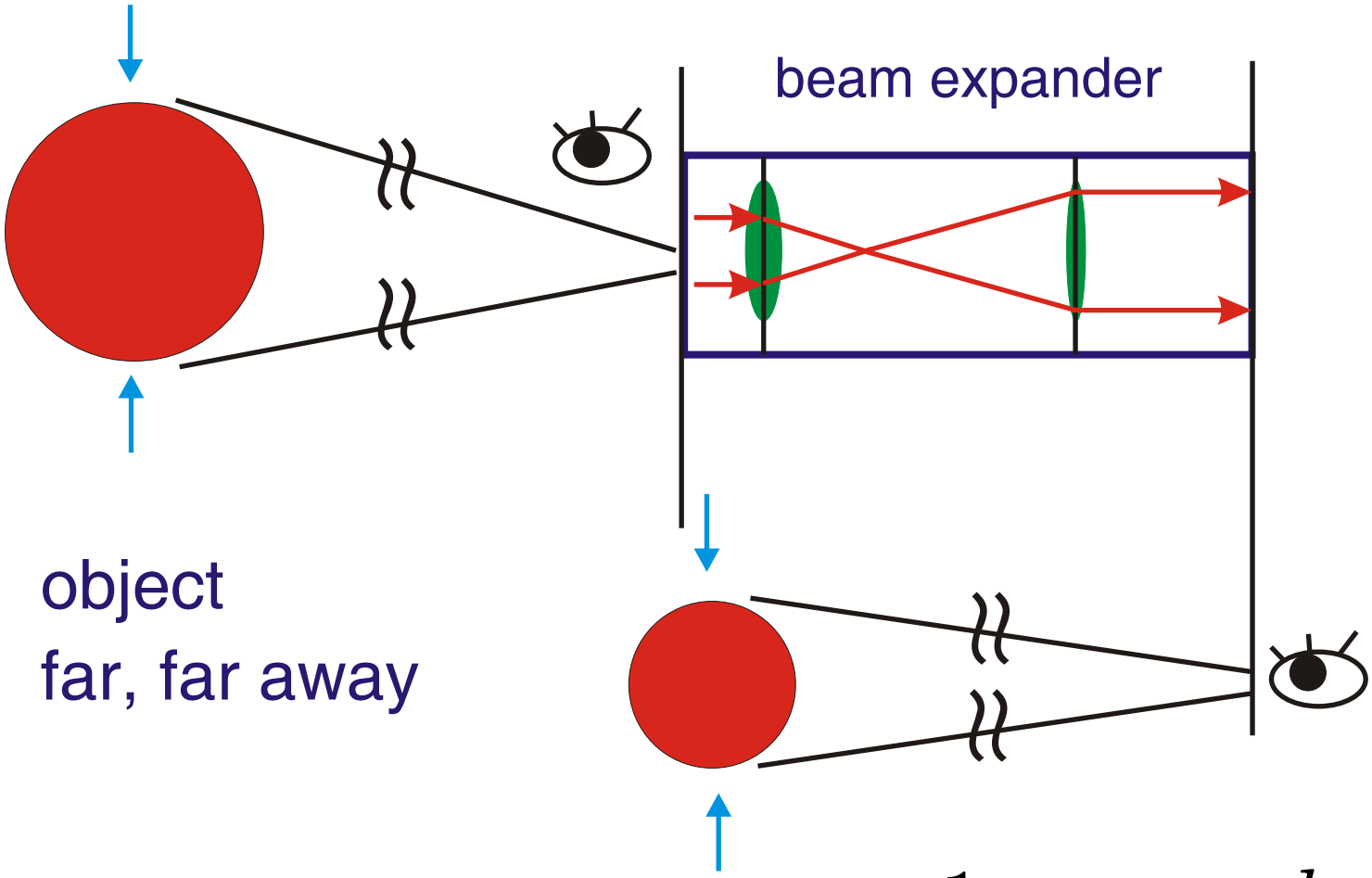



$$h_2 = -2h_1 + 3f_1\theta \simeq -2h_2$$

$$\theta_2 = -\theta_1/2$$

# Combination of Ray Matrix

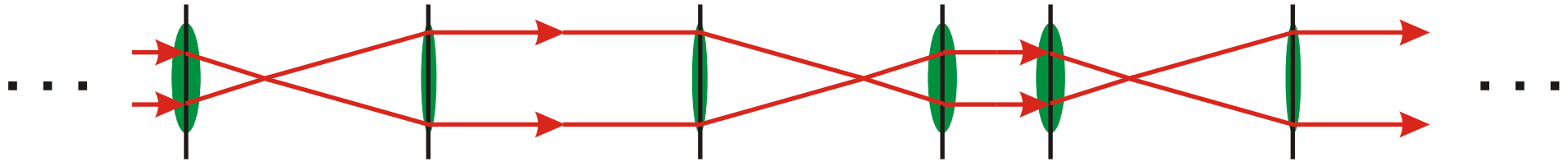
$$d_0 = R\theta_1$$



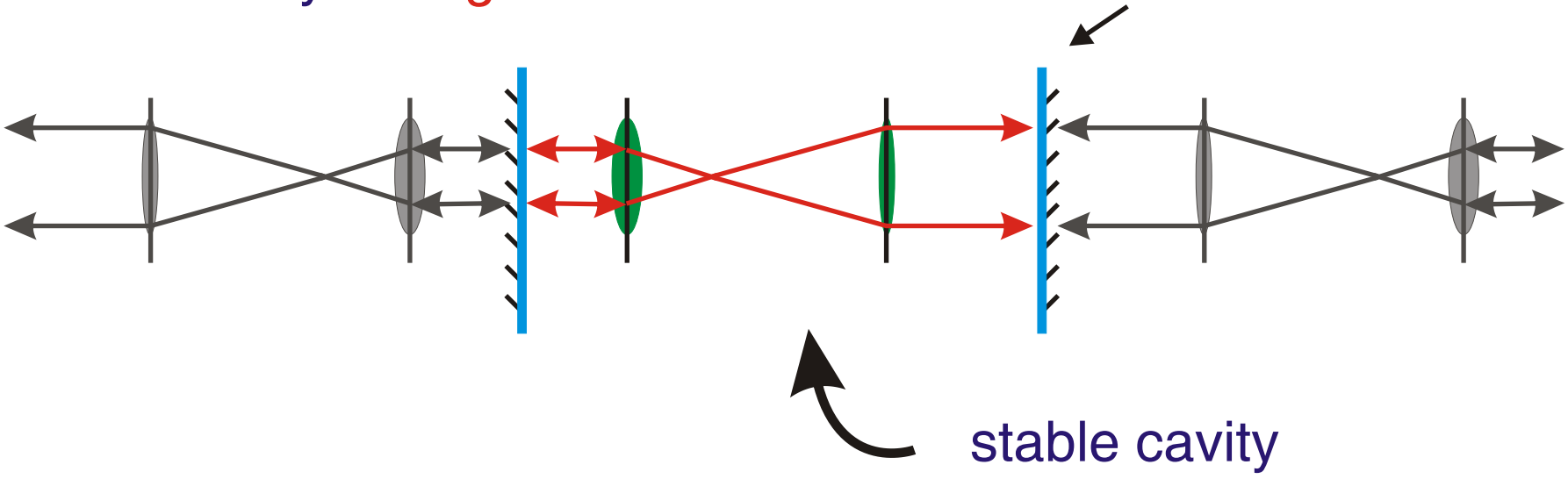

$$d = (R + 3f_1\theta_1)\theta_2 \simeq R\theta_2 = \frac{1}{2}R\theta_1 = \frac{d_0}{2}$$

# Application of Ray Matrix

ex. cavity analysis (periodical optical system)



eventually divergent or stable



# Guassian optics

Helmholtz equation

$$\nabla U(\mathbf{r}) + k^2 U(\mathbf{r}) = 0$$

with paraxial approx.

$$U(\mathbf{r}) = A(\mathbf{r}) e^{-jkz}$$

$$\rightarrow \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) A(\mathbf{r}) - jk \frac{\partial A(\mathbf{r})}{\partial z} = 0$$

paraxial Helmholtz eq.

Solution:

$$A(\mathbf{r}) = \frac{A_1}{q(z)} \exp\left( -jk \frac{x^2 + y^2}{2q(z)} \right)$$

q-parameter

$$\frac{1}{q(z)} = \frac{1}{R(z)} - j \frac{\lambda}{\pi w^2(z)}$$

q is complex.

# Guassian optics

## beam waist

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_0}\right)^2} \qquad w_0 = \sqrt{\frac{\lambda z_0}{\pi}}$$

## radius of curvature

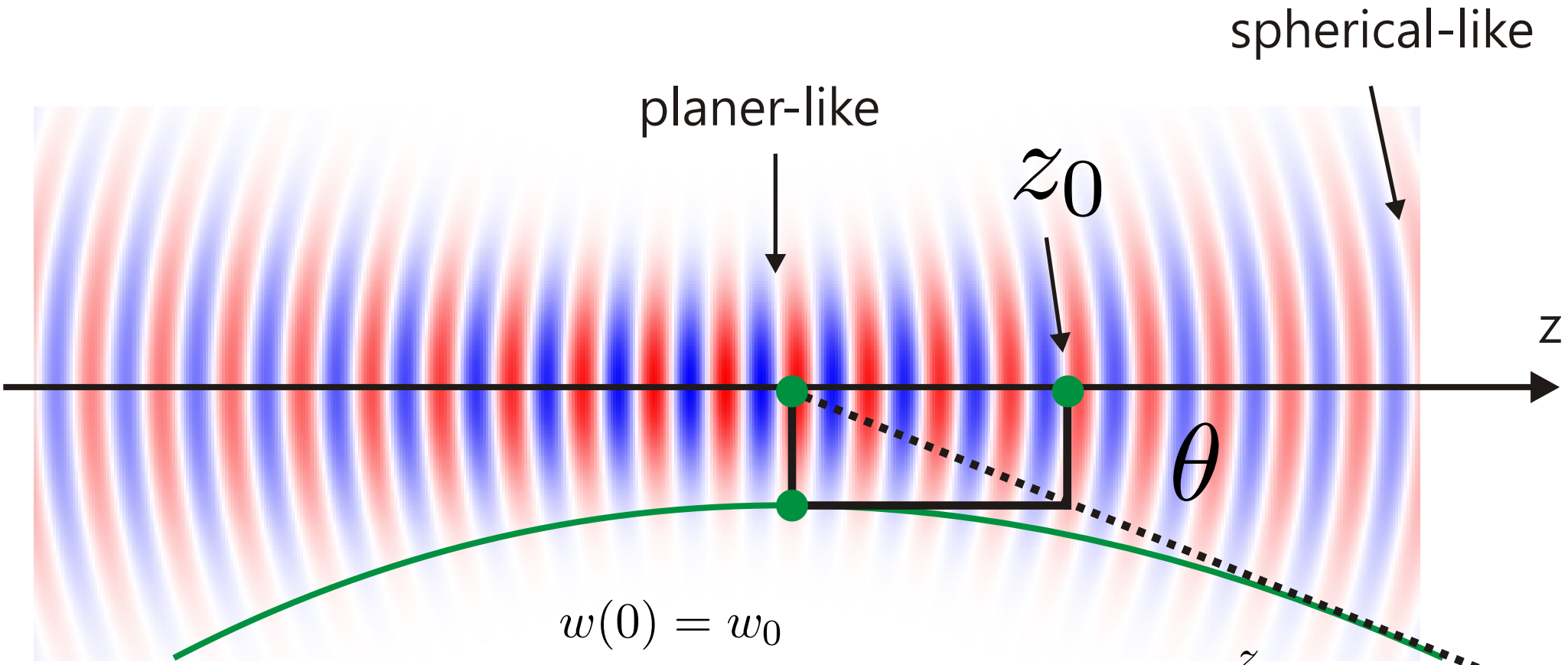
$$R(z) = z\left[1+\left(\frac{z_0}{z}\right)^2\right]$$

## divergence angle

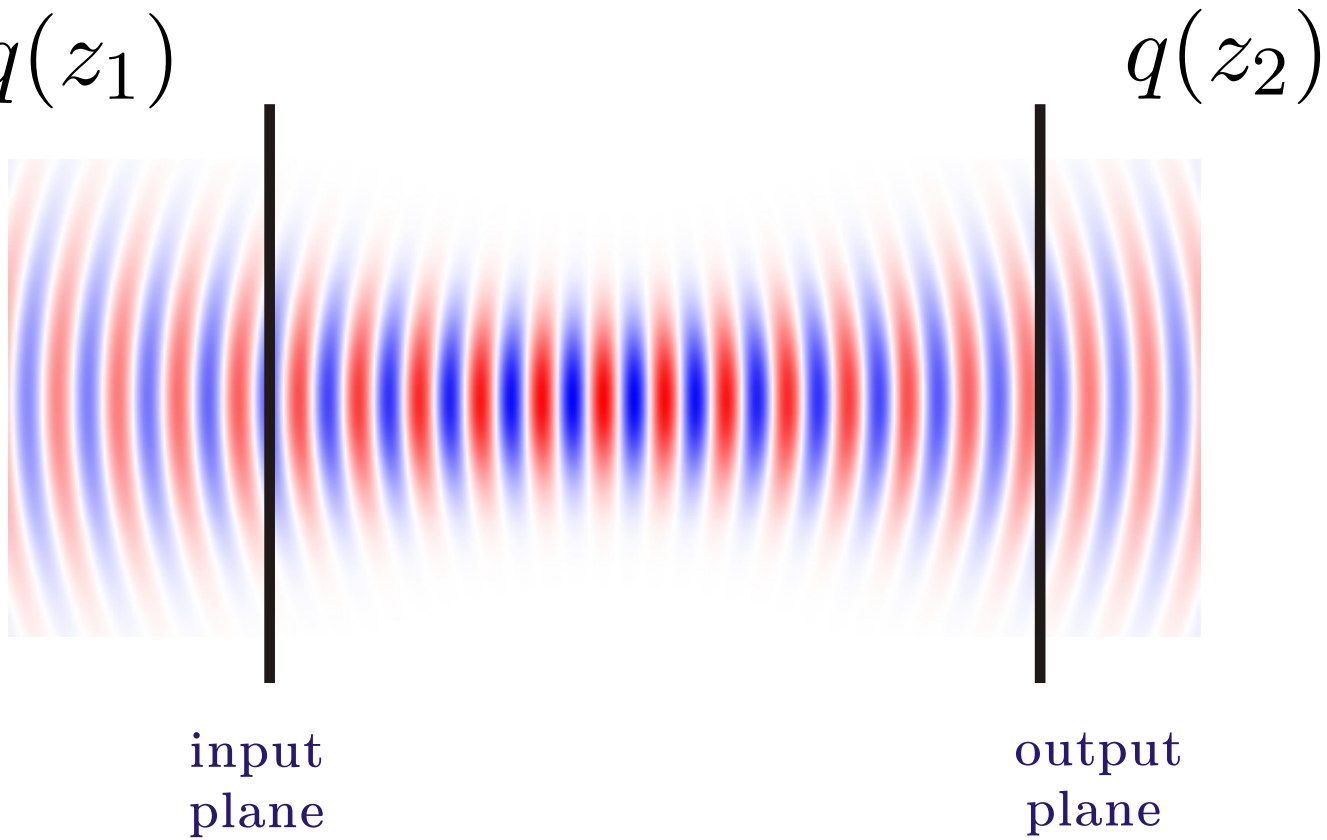
$$\theta = \frac{w_0}{z_0} = \frac{\lambda}{\pi w_0}$$

Rayleigh lengh $z_0$

# Guassian optics



# Propagation of guassian beam

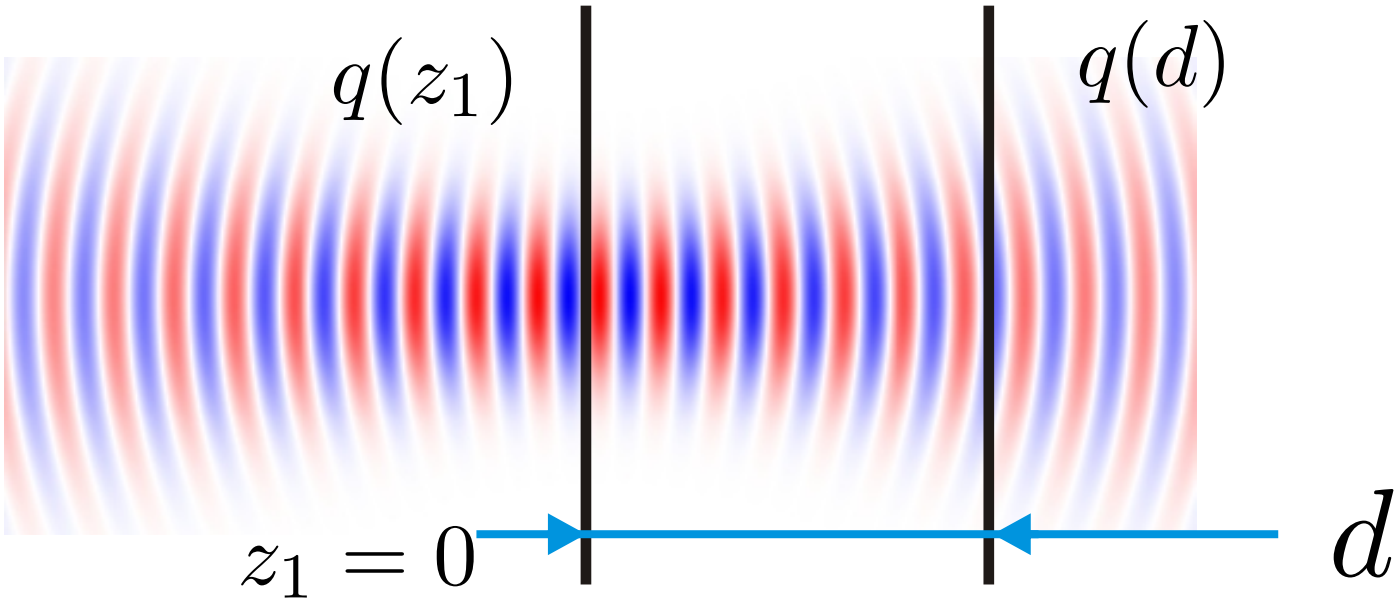



$$q(z_2) = \frac{Aq(z_1) + B}{Cq(z_1) + D}$$

ABCD matrix is the same as ray matrix

# q parameter with ABCD matrix

ex. propagates since focal spot in free space




$$\frac{1}{q_1} = \frac{1}{\infty} - j\frac{\lambda}{\pi w_0^2} \qquad q_1 = j\frac{\pi w_0^2}{\lambda}$$

# q parameter with ABCD matrix

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix} \qquad q_2 = \frac{q_1 + d}{0 \cdot q_1 + d} = d + j\frac{\pi w_0^2}{\lambda}$$

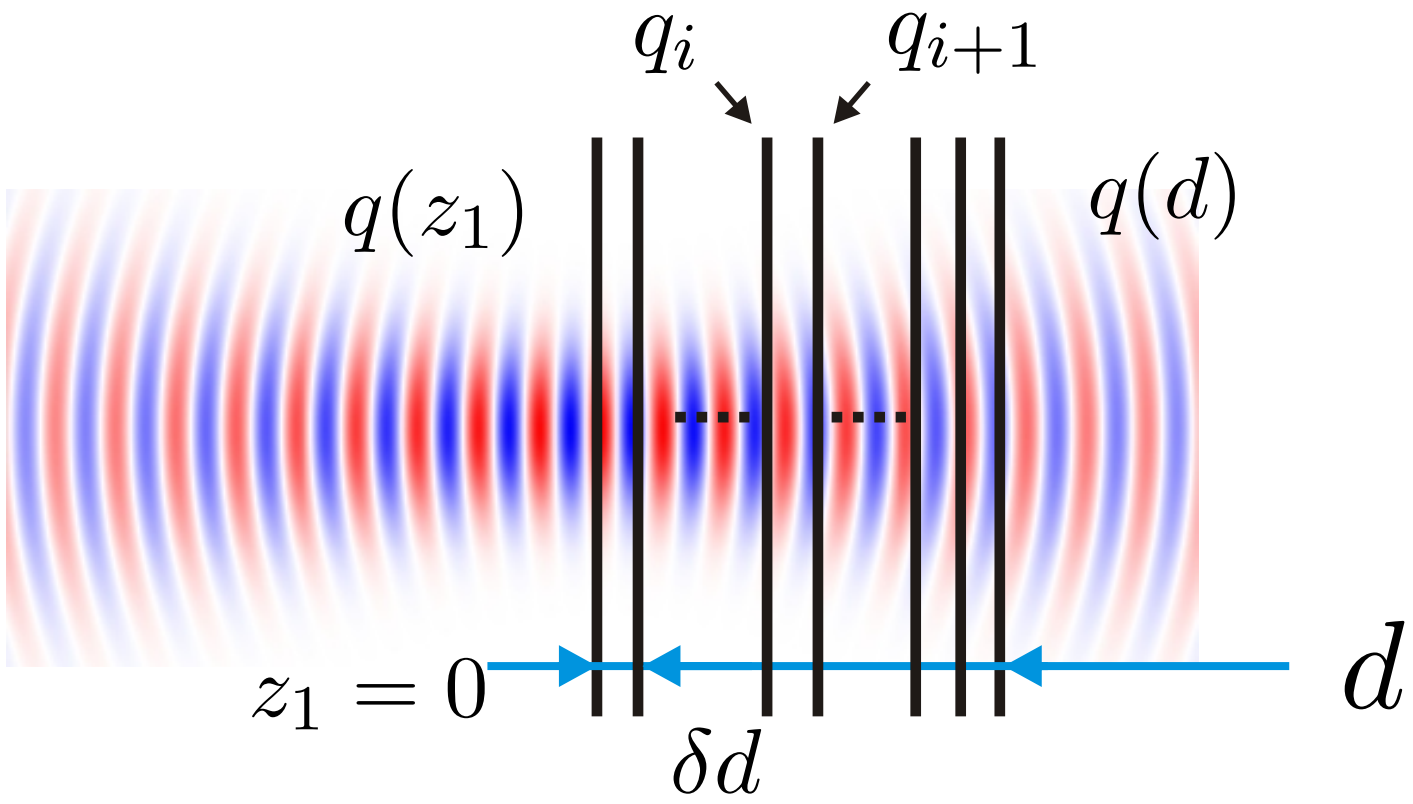
$$\frac{1}{q_2} = \left(d^2 + \frac{\pi^2 w_0^4}{\lambda^2}\right)\left(d - j\frac{\pi w_0^2}{\lambda}\right) = \frac{1}{R(d)} - j\frac{\lambda}{\pi w^2(d)}$$

$$R(d) = d\left[1 + \left(\frac{\pi w_0^2}{d\,\lambda}\right)^2\right]$$

$$w(d) = w_0\sqrt{1 + \left(\frac{d\,\lambda}{\pi w_0^2}\right)^2}$$

# q parameter with ABCD matrix

## ex. numerical solusion



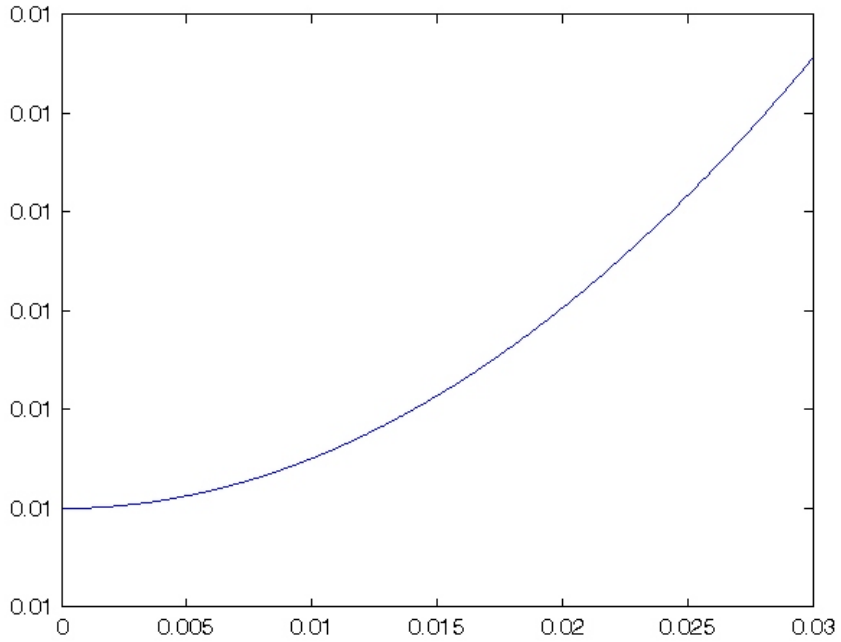

$$\sum_i \delta d = d$$

$$q_{i+1} = \frac{q_i + \delta d}{0 \cdot q_i + 1} = q_i + \delta d$$

# q parameter with ABCD matrix

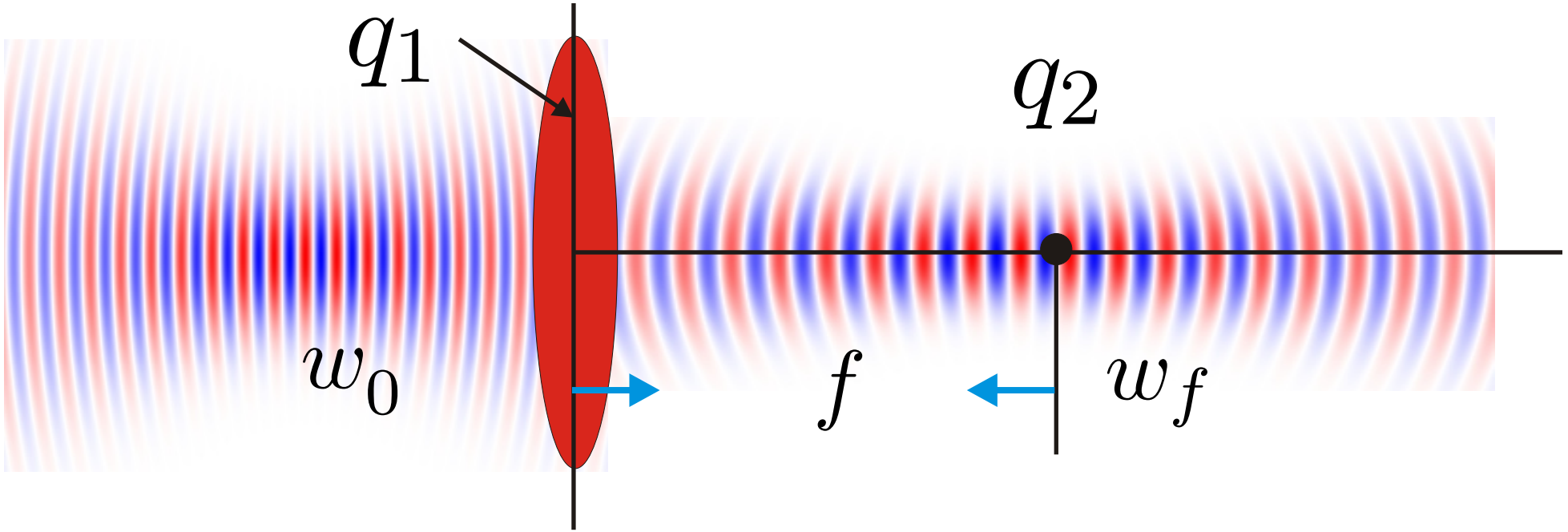
ex. matlab code

```
dd=3E-5;
step=1000;
R0=1E20;
w0=1E-2;
lambda=1E-6;
q0=1/(1/R0-j*lambda/(pi*w0^2));
qi=zeros(step+1,3)
%qi(step,1)=distance,
%qi(step,3)=q-parameter,
%zero(step,3)=waist
qi(1,1)=0;
qi(1,2)=q0;
qi(1,3)=w0;
for i=1:step
   qi(i+1,1)=qi(i,1)+dd;
   qi(i+1,2)=qi(i,2)+dd;
   qi(i+1,3)=sqrt(1/(-imag(1/qi(i+1,2)))*lambda/pi);
end
plot(qi(:,1),qi(:,3));
```



# q parameter with ABCD matrix

ex. focusing guassian beam




$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & f \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix} = \begin{bmatrix} 0 & f \\ -\frac{1}{f} & 1 \end{bmatrix}$$

# q parameter with ABCD matrix

$$q_2 = \frac{q_1 + f}{-q_1/f + 1} = \frac{j\frac{\lambda}{\pi w_0^2} + f}{-j\frac{\lambda}{\pi w_0^2} + 1}$$

$$\frac{\lambda}{\pi w_f^2} = -\left[1 + \frac{\lambda^2}{f^2\pi^2 w_0^4}\right] \frac{\lambda}{\pi w_0^2}$$

$w_f$ increases with $f$

# q parameter with ABCD matrix

## ex. numerical calculation with Matlab

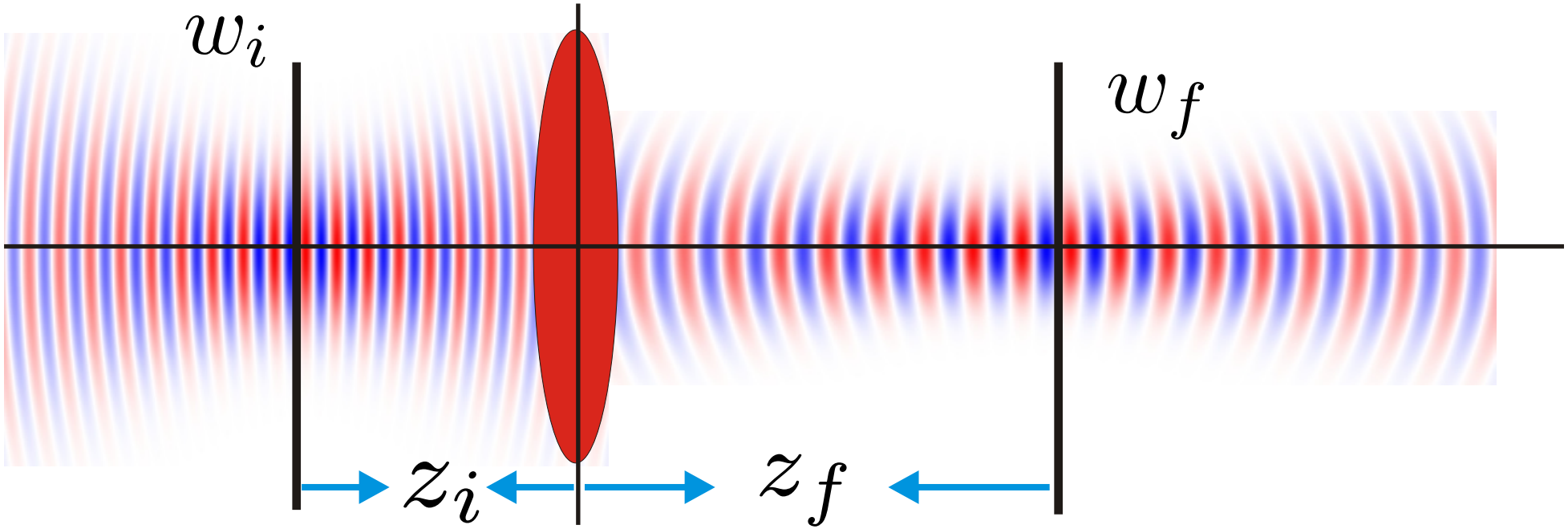
```
function wf=gb2(w0,f)
R0=1E10;
lambda=1E-6; % 1um
q0=1/(1/R0-j*lambda/(pi*w0^2));
qf=f/(-q0/f+1);
wf=sqrt(1/(-imag(1/qf))*lambda/pi);
% assum that beam wast @ lens is 5 mm

>> gb(5E-3, 15E-2)    % focusing with f= 15 cm len
ans = 4.776 E -006    % ~4.8 um

>>gb(5E-3,30E-2 )    % focusing with f= 30 cm len
ans = 9.5493 E-006  % ~9.5 um
```

# q parameter with ABCD matrix

ex. characteristic of lens transformation




$$w_i\theta_i = w_f\theta_f$$

$$\frac{1}{z_i} + \frac{1}{z_f} = \frac{1}{f}$$